

Continuous Gaussian Hidden Markov Models for Equity and Volatility Index Dynamics: A Baum-Welch Approach Without Jump Augmentation, with Cross-Asset SIM and Copula Extensions

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Abstract

We present a continuous hidden Markov model (CHMM) with Gaussian emissions trained via the Baum-Welch (Expectation-Maximization) algorithm for generating synthetic financial time series that preserve the canonical stylized facts of asset returns. Unlike the discrete-state framework of Alswaidan and Varner [1], which requires Laplace quantile binning and a Poisson jump-duration mechanism to achieve volatility clustering, we show that the continuous model at a single moderate state resolution ($K = 18$, selected from a sweep over $K \in \{3, 6, 9, 12, 15, 18, 21\}$) reproduces all three stylized facts (heavy tails, negligible linear autocorrelation, and persistent volatility clustering) without jump augmentation, thereby eliminating two tunable hyperparameters (ϵ, λ) and the discretization step entirely.

We benchmark the CHMM against six alternative generators on ten years of daily S&P 500 ETF (SPY) data: bootstrap resampling, Gaussian and Laplace i.i.d. generators, the *discrete* HMM of Alswaidan and Varner [1] in both no-jump (NJ) and Poisson-jump (WJ, $\epsilon = 0.01, \lambda = 3$) variants, and GARCH(1,1). Evaluation uses seven complementary metrics (Kolmogorov–Smirnov and Anderson–Darling pass rates, excess kurtosis, autocorrelation function mean absolute error, Wasserstein-1 and Hellinger distances, and quantile coverage) applied to 1,000 simulated paths in both in-sample (2014–2024) and out-of-sample (2025) windows. Cross-asset generalization to NVDA, JNJ, JPM, AAPL, and QQQ confirms adaptability to diverse risk profiles.

To move from univariate to multi-asset generation we extend the framework with two cross-asset dependence models that preserve each asset’s fitted CHMM marginal: a Single Index Model (SIM) factor construction with SPY as the market engine, and Gaussian / Student-t copulas fitted to rank-based correlation and simulated via rank reordering. On six SPY-member assets the copula-based dependence models substantially outperform the SIM factor baseline on per-asset distributional accuracy (median in-sample KS pass rate 98.0% for both copulas versus 83.8% for SIM) while also delivering closer correlation-matrix reproduction (off-diagonal MAE 0.027 for Student-t, 0.031 for Gaussian, and 0.078 for SIM). The Student-t copula’s profile MLE selected $\nu^* = 6$ degrees of freedom, indicating economically meaningful joint tail dependence.

We further extend the framework to CBOE Volatility Index (VIX) and VXX log returns, demonstrating that the CHMM captures the distinct statistical properties of volatility indices (including stronger mean reversion, positive skewness, and pronounced regime structure). The VIX regime decomposition provides a natural bridge to option pricing: median VIX levels per decoded regime define a volatility map that drives regime-switching geometric

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Brownian motion paths. A parametric stochastic-volatility comparison against the resulting implied volatility surface is the subject of companion work and is deliberately outside the scope of this paper.

Keywords: Continuous Hidden Markov Model, Baum-Welch Algorithm, Stylized Facts, Volatility Clustering, Regime-Switching, Synthetic Financial Data, Single Index Model, Copula Dependence, Cross-Asset Simulation, VIX.

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance. High-fidelity synthetic data are essential for stress testing risk models against plausible but unobserved market scenarios, validating portfolio optimization algorithms, and augmenting limited training sets for machine learning pipelines [2, 3]. Empirical equity excess growth rates exhibit three well-documented statistical regularities, namely heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in raw returns, and persistent volatility clustering, that any generative framework must simultaneously reproduce to be considered statistically faithful [4, 5].

Parametric approaches to synthetic data generation face a fundamental tension between distributional accuracy and temporal fidelity. GARCH models [6, 7] excel at reproducing the slow decay of the autocorrelation function (ACF) of absolute returns, the hallmark of volatility clustering, but their unimodal innovation distribution cannot represent the multi-regime structure of equity markets, leading to poor distributional fit. Conversely, i.i.d. generators that match the marginal distribution (for example, Laplace sampling) achieve high distributional pass rates but produce flat ACF profiles by construction. Hidden Markov models (HMMs) offer a middle path: the latent regime structure can simultaneously generate heavy-tailed mixtures and temporal persistence through the transition dynamics.

However, the application of HMMs to financial stylized facts has been constrained by an influential negative result. Rydén et al. [8] systematically evaluated Gaussian-emission HMMs with $K = 2\text{--}3$ states against the canonical stylized facts and concluded that while HMMs reproduce heavy tails and negligible return autocorrelation, they *fail* to capture the slow decay of the ACF of absolute returns. The mechanism is clear: with few states, the geometric sojourn-time distribution of a standard Markov chain produces exponentially fast ACF decay, insufficient to match the hyperbolic-like persistence observed empirically. Bulla and Bulla [9] subsequently showed that hidden semi-Markov models (HSMs), which replace the geometric sojourn distribution with flexible alternatives, restore ACF fidelity at the cost of substantially increased model complexity.

Alswaidan and Varner [1] proposed an alternative solution within the discrete HMM framework: a Poisson jump-duration mechanism that forces the model to dwell in tail states for empirically calibrated durations. That approach achieved 91–95% Kolmogorov–Smirnov (KS) pass rates across 424 US equities and partially reproduced volatility clustering. However, the discrete framework introduced three sources of complexity. First, Laplace quantile binning discretizes continuous observations and introduces quantization error. Second, two additional hyperparameters (ϵ for jump probability and λ for mean jump duration) require grid-search calibration. Third, within-bin Student- t emission models are estimated separately from the transition structure.

In this paper, we show that these complexities are unnecessary. A continuous Gaussian HMM (CHMM) trained via the Baum-Welch algorithm [10, 11] with quantile-based initialization reproduces all three canonical stylized facts at moderate state resolution *without* requiring jump augmentation, discretization, or separate emission fitting. We sweep the state resolution over $K \in \{3, 6, 9, 12, 15, 18, 21\}$, report the full sensitivity in the appendix, and select $K = 18$ as the single main-body benchmark on the basis of a multi-metric score: $K = 18$ is the only value that

wins (or ties) on all four distributional pass-rate metrics (IS/OoS KS and IS/OoS AD) while also attaining the smallest Wasserstein-1 error and the largest simulated kurtosis across the sweep. The key insight is that the negative result of Rydén et al. [8] is an artifact of insufficient state resolution: at $K \geq 3$ with proper initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to generate slow ACF decay, effectively approximating the flexible sojourn-time distributions of HSMMs through a mixture of geometric distributions.

Beyond single-asset fidelity, generating *jointly plausible* multi-asset scenarios is essential for portfolio risk applications, yet it is not directly addressed by the univariate HMM literature. A factor-model construction such as the Single Index Model (SIM) of Sharpe [12] propagates a market engine to a cross-section, but at the cost of imposing a rigid linear dependence structure and replacing each asset’s carefully fitted marginal with the regression fit. Copula theory [13, 14] provides a conceptually cleaner decomposition: a joint distribution factors uniquely into its marginals and a copula that encodes dependence, so the per-asset CHMM can be reused verbatim as the marginal while a Gaussian or Student-t copula [15, 16] injects cross-sectional co-movement. We implement both constructions on a representative six-asset subset (SPY, NVDA, JNJ, JPM, AAPL, QQQ) and report a consistent ordering in which copulas outperform the SIM factor baseline both on per-asset marginal fidelity and on correlation-matrix reproduction.

The contributions of this work are:

1. **Baum-Welch training without jump augmentation.** We train continuous Gaussian emission parameters (μ_k, σ_k) and the transition matrix $\mathbf{T}$ jointly from raw observations via the Baum-Welch algorithm with quantile-based initialization. At $K = 18$, the resulting CHMM achieves 96.8% IS and 95.0% OoS KS pass rates, AD pass rates of 90.1% IS and 87.2% OoS, and ACF-MAE of 0.051 on absolute returns, all without any jump mechanism, and eliminating the two hyperparameters (ϵ, λ) and the discretization step.
2. **Contradiction of the Rydén et al. (1998) consensus.** We demonstrate that the widely cited limitation of Gaussian HMMs, the inability to reproduce volatility clustering, is a consequence of testing only $K = 2\text{--}3$ states. At moderate K , the learned transition matrix develops sufficient persistence in high-volatility states to achieve ACF-MAE comparable to GARCH, which we show is effectively an approximation to the HSMM solution of Bulla and Bulla [9].
3. **Seven-model benchmark including the discrete HMM of the prior paper.** We evaluate the CHMM against six alternatives on identical metrics and identical data: bootstrap resampling, Gaussian and Laplace i.i.d. generators, the *discrete* HMM of Alswaidan and Varner [1] in both no-jump (NJ) and Poisson-jump (WJ, $\epsilon = 0.01$, $\lambda = 3$) variants, and GARCH(1,1). Seven complementary metrics (KS and Anderson–Darling pass rates, excess kurtosis, ACF-MAE, Wasserstein-1 and Hellinger distances, and quantile coverage) are reported on 1,000 simulated paths in both in-sample (2014–2024) and out-of-sample (2025) windows. The direct discrete-vs-continuous comparison isolates the contribution of our methodological changes relative to the prior paper.
4. **State resolution analysis with principled model selection.** We systematically vary $K \in \{3, 6, 9, 12, 15, 18, 21\}$ and report all seven metrics per K . Distributional fidelity improves with K and peaks at $K = 18$ across IS and OoS KS and AD simultaneously; ACF-MAE is remarkably stable across all K (0.046–0.051) and is lowest at $K = 3$, crystallizing a distributional–temporal tradeoff that we make explicit.
5. **Cross-asset SIM and copula extensions.** We extend the univariate framework to multi-asset synthesis through two complementary cross-sectional dependence models that both preserve each asset’s fitted CHMM marginal: a Single Index Model with the CHMM-driven SPY path as the market engine, and Gaussian / Student-t copulas fitted by Kendall’s

τ and simulated via rank reordering. On six representative assets, copulas substantially outperform the SIM factor baseline on per-asset in-sample KS pass rate (median 98.0% versus 83.8%) and on the off-diagonal correlation MAE (0.027 for Student-t, 0.031 for Gaussian, 0.078 for SIM), with the Student-t copula profile MLE selecting $\nu^* = 6$ degrees of freedom.

6. **Volatility index generalization and regime-switching volatility for option pricing.** We confirm robustness across NVDA, JNJ, JPM, AAPL, and QQQ, and extend the framework to CBOE VIX and VXX log returns, demonstrating that the CHMM captures the distinct statistical properties of volatility indices and provides a natural bridge to regime-switching option pricing through the decoded per-regime volatility map. Benchmarking the resulting implied-volatility surface against a parametric stochastic-volatility model is the subject of a companion paper and is intentionally outside the scope of this study.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on stylized facts, parametric models, HMMs, cross-asset dependence, and deep generative approaches. Section 3 describes the data, the CHMM specification, the Baum-Welch training procedure, the benchmark models, the cross-asset SIM and copula constructions, and the validation metrics. Section 4 presents the empirical results: descriptive statistics, seven-model comparison at $K = 18$, a summary state resolution sensitivity analysis, out-of-sample evaluation, cross-asset SIM and copula dependence, and the VIX/VXX extension. Section 5 interprets the findings, and Section 6 summarizes the contributions and outlines future directions.

2 Related Work

Research on reproducing the canonical stylized facts of financial markets spans parametric return models, latent-state frameworks, multi-asset dependence modeling, and deep generative approaches. This section reviews the key developments that motivate the present work, with particular attention to the Gaussian HMM limitation identified by Rydén et al. [8] that our results challenge, and to the cross-asset dependence models that we use in our multi-asset extension.

2.1 Stylized Facts of Asset Returns

The Black–Scholes–Merton framework [17] assumes that log-returns are independently and identically distributed (i.i.d.) Gaussian random variables. This assumption, while analytically convenient, is contradicted by a large body of empirical evidence accumulated over six decades. Mandelbrot [5] demonstrated that speculative price changes exhibit heavy tails (leptokurtosis), with extreme returns occurring far more frequently than a Gaussian model predicts. Fama [18] confirmed negligible linear autocorrelation in raw returns, consistent with informational efficiency, but this does not imply independence: nonlinear dependence persists in the form of volatility clustering. Schwert [19] documented that the autocorrelation of absolute returns decays slowly, often following a hyperbolic rather than exponential pattern, providing the empirical signature of clustered volatility.

Cont [4] systematized these observations into a canonical set of stylized facts that hold across asset classes, geographic markets, and sampling frequencies:

1. *Heavy tails:* the marginal distribution of returns exhibits positive excess kurtosis, with power-law tail decay.
2. *Absence of linear autocorrelation:* raw returns are approximately uncorrelated, consistent with market efficiency.

3. *Volatility clustering*: the autocorrelation of absolute (or squared) returns is positive and decays slowly, indicating that large price movements tend to be followed by large movements of either sign.

Any generative model for financial time series must simultaneously reproduce all three facts to be considered statistically faithful. Failure to capture even one, as we will show for the Gaussian i.i.d. generator (distributional failure) and GARCH(1,1) (distributional failure despite temporal fidelity), renders the synthetic data inadequate for downstream applications such as risk measurement and portfolio backtesting.

2.2 Parametric Volatility Models

The ARCH model of Engle [6] introduced conditional heteroscedasticity into time series econometrics, modeling the variance of returns as a function of past squared residuals. The generalized ARCH (GARCH) extension of Bollerslev [7] added lagged conditional variance, producing the workhorse GARCH(1,1) specification

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (1)$$

where $\alpha + \beta < 1$ ensures stationarity. The single-decay parameter β directly controls the persistence of the conditional variance process, producing exponential ACF decay that closely approximates the empirical pattern when $\alpha + \beta$ is near unity. However, GARCH models assume a unimodal innovation distribution and cannot represent discrete market regimes. As we demonstrate empirically, this unimodal assumption leads to catastrophic distributional failure: GARCH(1,1) achieves only 13% KS pass rate despite matching the observed ACF-MAE almost exactly (0.047 vs. 0.048 observed).

Parametric stochastic volatility models treat log-variance as a latent diffusion and produce closed-form or semi-analytic option prices, but parameter estimation requires computationally intensive methods such as particle filtering or Markov-chain Monte Carlo. A detailed comparison of the CHMM-implied volatility surface against a parametric stochastic-volatility benchmark is the subject of companion work and is not revisited here.

Jump-diffusion models [20, 21] augment the geometric Brownian motion with Poisson-driven jumps in the price level, producing heavy-tailed marginal distributions. Merton [20] used Gaussian jump sizes, while Kou [21] introduced double-exponential jumps to capture the asymmetry between upside and downside tail risk. These models generate heavy tails but lack a mechanism for volatility persistence: after a jump event, the volatility returns immediately to its baseline level, failing to reproduce the clustering phenomenon.

2.3 Hidden Markov Models and the Rydén et al. Limitation

Hamilton [22] introduced the Markov-switching model to economics, demonstrating that U.S. GNP growth is well described by a two-state regime-switching process. The model posits a discrete latent state S_t that evolves as a first-order Markov chain, with the observable variable drawn from a state-dependent distribution. HMMs have since been applied to financial time series for regime identification [23], volatility estimation [24], and trading strategy design [25].

The seminal evaluation of HMMs against stylized facts is due to Rydén et al. [8], who fitted Gaussian-emission HMMs with $K = 2$ and $K = 3$ states to daily Swedish equity returns. They established three key findings. First, HMMs produce heavy-tailed marginal distributions through the Gaussian mixture mechanism, even though each component is thin-tailed. Second, HMMs produce negligible linear autocorrelation in returns, consistent with the observed stylized fact. Third, HMMs *fail* to reproduce the slow decay of the ACF of absolute returns (volatility clustering). The third finding follows from a structural property of Markov chains: the sojourn time in any state follows a geometric distribution, which implies exponential decay of the ACF.

For $K = 2-3$ states, the exponential decay is too fast to approximate the hyperbolic-like persistence observed empirically. This result has been widely cited as a fundamental limitation of standard HMMs for financial modeling.

Bulla and Bulla [9] proposed hidden semi-Markov models (HSMMs) as a remedy. HSMMs replace the geometric sojourn-time distribution with flexible parametric alternatives such as negative binomial or log-series, allowing arbitrarily slow ACF decay. They demonstrated that HSMMs with $K = 2-4$ states and negative-binomial sojourn distributions successfully reproduce all three stylized facts, confirming that the Rydén limitation is fundamentally about the sojourn-time distribution, not the HMM framework itself.

The present work offers an alternative resolution: rather than modifying the sojourn-time distribution, we increase the state resolution. With $K \geq 3$ states and quantile-based initialization, the Baum-Welch algorithm learns a transition matrix where high-volatility states exhibit sufficient self-transition probability (diagonal dominance) to sustain persistence. The resulting mixture of K geometric sojourn-time distributions, each with a different rate parameter, can approximate slow ACF decay, effectively achieving an HSMM-like effect within the standard HMM framework. This insight has not been explored in the prior literature, likely because Rydén et al. [8] only tested $K = 2-3$ and subsequent work adopted the HSMM path.

2.4 Discrete HMM with Jump Augmentation

Alswaidan and Varner [1] addressed the Rydén limitation through a different mechanism: a Poisson jump-duration process that forces the model to dwell in tail states for durations drawn from a Poisson distribution, overriding the geometric sojourn time. Their discrete framework used Laplace quantile binning to partition continuous observations into categorical states, estimated the transition matrix by direct frequency counting, and employed Student- t within-bin emission distributions. The jump mechanism introduced two hyperparameters (ϵ for jump probability, λ for mean duration) calibrated via grid search. The framework achieved 91–95% KS pass rates across 424 US equities and partially reproduced volatility clustering.

The present work eliminates all three sources of complexity introduced by the discrete framework. First, continuous Gaussian emissions replace Laplace quantile binning, removing discretization artifacts. Second, the Baum-Welch algorithm jointly learns emissions and transitions, replacing frequency counting. Third, no jump mechanism is needed at moderate K , removing two hyperparameters and the grid search entirely.

2.5 Cross-Asset Dependence Models

Generating jointly plausible multi-asset paths is a separate problem from univariate fidelity. Two approaches dominate the quantitative finance literature.

Single Index Model (SIM). Sharpe [12] proposed a one-factor decomposition in which each asset’s return is an affine function of a single market factor plus an idiosyncratic residual, $G_{i,t} = \alpha_i + \beta_i G_{M,t} + \eta_{i,t}$. The SIM is parsimonious, analytically tractable, and is the workhorse reduction behind the mean-variance portfolio construction of Markowitz [26]. Its main limitations for synthetic data generation are that (i) the linear affine map distorts each asset’s marginal distribution, so per-asset distributional fidelity degrades relative to the stand-alone marginal model, and (ii) it imposes a rank-one correlation structure in which all cross-asset dependence is mediated by the market factor.

Copulas. Sklar’s theorem [13, 14] states that any joint cumulative distribution H with continuous marginals $F_1, \dots, F_d$ factors uniquely as

$$H(g_1, \dots, g_d) = C(F_1(g_1), \dots, F_d(g_d)), \quad (2)$$

where the copula $C : [0, 1]^d \rightarrow [0, 1]$ is a multivariate distribution with uniform margins. This factorization allows the joint distribution to be assembled from arbitrary per-asset marginals and

any dependence structure C . Within quantitative finance, the Gaussian and Student-t copulas [15, 16] are standard choices because they admit analytic densities, are parameterized by a correlation matrix, and (for the Student-t copula) admit non-zero symmetric tail dependence controlled by the degrees-of-freedom parameter ν . Simulation via rank reordering [27] exploits the uniqueness of Sklar’s decomposition to inject the copula-sampled dependence structure onto independently simulated marginals without distorting any marginal distribution.

We adopt the copula construction of [16] for our cross-asset extension: per-asset CHMM simulations provide the marginals, Kendall’s τ provides a robust rank-based correlation estimator, and the Student-t copula’s degrees of freedom are selected by profile maximum likelihood. The empirical ordering we report in Section 4.7 (copulas ahead of SIM on both per-asset KS and cross-asset correlation reproduction) matches the ordering reported by Alswaidan and Varner [1] for the discrete HMM marginals, suggesting that the copula advantage is intrinsic to the rank-reordering construction rather than specific to the marginal model.

2.6 Deep Generative Models

Deep generative models have attracted growing interest for financial time series synthesis. Takahashi et al. [28] applied generative adversarial networks (GANs) to stock return generation, finding that GAN-produced sequences visually resembled financial data but failed to reproduce ACF structure (the same volatility-clustering limitation as low- K HMMs). Kwon and Lee [29] systematically evaluated several GAN architectures against the Cont stylized facts and confirmed that volatility clustering remained the most difficult property to capture, unless the architecture was specifically designed to encode temporal dependence (for example, through recurrent discriminators).

Alswaidan and Varner [1] benchmarked a GRU-based generator against HMM variants and found that it closed 83% of the ACF-MAE gap relative to the HMM, but suffered variance collapse (0.6% KS pass rate), indicating that distributional fidelity and temporal structure remain in tension for neural approaches. The present study focuses on the parametric HMM approach, which offers interpretability (each state corresponds to a named market regime), computational efficiency ($O(TK^2)$ per Baum-Welch iteration), and guaranteed convergence (EM monotonicity), advantages that complement the flexibility of deep generative methods.

2.7 Quality Assessment for Synthetic Time Series

Stenger et al. [30] surveyed the quality assessment landscape for synthetic time series, organizing evaluation metrics into distributional fidelity, temporal fidelity, and downstream utility categories. Our seven-metric evaluation framework (KS and AD pass rates, excess kurtosis, ACF-MAE, Wasserstein-1, Hellinger distance, and quantile coverage) spans the first two categories, providing a comprehensive assessment that avoids reliance on any single metric. As we will demonstrate, no single model dominates all seven metrics, and the choice of generator depends on which dimension of fidelity is prioritized. Assefa et al. [2] made a complementary argument in the specific context of financial data, emphasizing that synthetic generators should be evaluated on their ability to preserve statistical properties relevant to the intended downstream application.

3 Methods

3.1 Data

The empirical foundation of this study is a dataset of 424 United States-listed equities and exchange-traded funds spanning ten years of daily trading data (January 3, 2014 to December 31, 2024). We focus the primary single-asset analysis on the SPDR S&P 500 ETF Trust (SPY),

which tracks the S&P 500 index and is the most liquid equity ETF by trading volume. The data include daily open, high, low, and close prices along with volume and volume-weighted average price (VWAP) metrics, sourced from standard financial data providers. The in-sample (IS) dataset comprises $T_{\text{IS}} = 2,766$ trading days. A subsequent $T_{\text{OoS}} = 219$ trading days in 2025 are reserved as a held-out out-of-sample (OoS) test set to evaluate generalization.

Cross-asset generalization is assessed on five additional tickers spanning distinct risk profiles: NVDA (NVIDIA Corporation; high-beta technology), JNJ (Johnson & Johnson; low-beta healthcare), JPM (JPMorgan Chase & Co.; moderate-beta financials), AAPL (Apple Inc.; large-cap consumer technology), and QQQ (Invesco QQQ Trust; Nasdaq-100 ETF). For the univariate fidelity evaluation each asset is fitted independently using the same training/test split. For the cross-asset extension of Section 3.8 we additionally stack the six-asset in-sample return matrix $\mathbf{R}_{\text{IS}} \in \mathbb{R}^{T_{\text{IS}} \times 6}$ to fit the cross-sectional dependence models.

To extend the framework beyond equity returns, we additionally apply it to the CBOE Volatility Index (VIX) [31], which measures the 30-day implied volatility of S&P 500 options and is widely regarded as the market’s “fear gauge.” VIX daily close prices spanning approximately 2006–2024 constitute the in-sample dataset, with 2025 data reserved for out-of-sample testing. VIX log returns exhibit statistical properties that differ markedly from equity returns, including positive skewness (reflecting asymmetric volatility spikes), pronounced mean reversion, and extreme excess kurtosis, making it a challenging test case that probes the generality of the CHMM framework.

3.2 Excess Growth Rate Calculation

We compute the annualized excess growth rate $G_{i,j}$ for ticker i between trading days $j - 1$ and j given the equity price series $P_{i,j}$:

$$G_{i,j} \equiv \frac{1}{\Delta t} \cdot \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f \quad (3)$$

where $\Delta t = 1/252$ (one trading day expressed in years) and r_f is a constant risk-free rate set to zero throughout this study. The resulting excess growth rate has units of year^{-1} and is time-additive under continuous compounding. The annualization by $1/\Delta t$ scales the daily log-returns to an annualized basis, producing values with magnitudes on the order of ± 2 standard deviations for typical daily moves.

3.3 Continuous Hidden Markov Model

We define the continuous hidden Markov model (CHMM) as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ [11], where

- $S_t \in \mathcal{S} = \{1, 2, \dots, K\}$ is the hidden state at time t , representing the prevailing market regime;
- $\mathcal{O} \subseteq \mathbb{R}$ is the continuous observation space of excess growth rates;
- $\mathbf{T} \in \mathbb{R}^{K \times K}$ is the state transition probability matrix with entries $T_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i)$, satisfying $\sum_{j=1}^K T_{ij} = 1$ for all i ;
- $\mathbf{B} = \{(\mu_k, \sigma_k)\}_{k=1}^K$ specifies the Gaussian emission distributions, where $G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$;
- $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ is the initial state distribution with $\pi_k = \mathbb{P}(S_1 = k)$.

The emission density for state k is

$$b_k(x) = \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right). \quad (4)$$

The marginal distribution of G_t under the stationary regime is a K -component Gaussian mixture,

$$f(x) = \sum_{k=1}^K \bar{\pi}_k b_k(x), \quad (5)$$

where $\bar{\pi}$ is the stationary distribution satisfying $\bar{\pi} = \bar{\pi}\mathbf{T}$. This mixture structure is the mechanism by which HMMs produce heavy-tailed marginal distributions from thin-tailed Gaussian components: tail states with large $|\mu_k|$ or σ_k contribute to the kurtosis of the mixture even though each component has zero excess kurtosis individually.

Unlike the discrete framework of Alswaidan and Varner [1], which discretizes observations into Laplace quantile bins and estimates $\mathbf{T}$ by frequency counting, we learn $\mathbf{T}$, $\{\mu_k\}_{k=1}^K$, $\{\sigma_k\}_{k=1}^K$, and π jointly from the continuous observations via the Baum-Welch algorithm described below.

3.4 Baum-Welch (EM) Algorithm

The Baum-Welch algorithm [10] is the standard expectation-maximization (EM) procedure for maximum likelihood estimation of HMM parameters. Given a sequence of T observations $O_{1:T} = (O_1, O_2, \dots, O_T)$, the algorithm iterates between an E-step and an M-step until convergence.

E-Step: Forward-Backward in Log-Space. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_1, O_2, \dots, O_t, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1}, O_{t+2}, \dots, O_T \mid S_t = k, \mathcal{M})$. The recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1) \quad (6)$$

$$\log \alpha_t(j) = \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij} + \log b_j(O_t)), \quad t = 2, \dots, T \quad (7)$$

$$\log \beta_T(k) = 0 \quad (8)$$

$$\log \beta_t(i) = \underset{j}{\text{logsumexp}}(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1 \quad (9)$$

where the logsumexp operation is defined as

$$\underset{i}{\text{logsumexp}}(x_i) = x_{\max} + \log\left(\sum_i \exp(x_i - x_{\max})\right), \quad x_{\max} = \max_i x_i. \quad (10)$$

All forward-backward computations are performed entirely in log-space to prevent the floating-point underflow that occurs when multiplying many small probabilities for long observation sequences [32].

The posterior state occupancy probability is

$$\gamma_t(k) = \mathbb{P}(S_t = k \mid O_{1:T}, \mathcal{M}) = \frac{\alpha_t(k) \beta_t(k)}{\sum_{j=1}^K \alpha_t(j) \beta_t(j)} \quad (11)$$

and the posterior transition probability is

$$\xi_t(i, j) = \mathbb{P}(S_t = i, S_{t+1} = j \mid O_{1:T}, \mathcal{M}) = \frac{\alpha_t(i) T_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{m=1}^K \sum_{n=1}^K \alpha_t(m) T_{mn} b_n(O_{t+1}) \beta_{t+1}(n)}. \quad (12)$$

M-Step: Parameter Re-Estimation. Given the posterior quantities $\gamma_t(k)$ and $\xi_t(i, j)$, the parameters are re-estimated as

$$\pi_k^{\text{new}} = \gamma_1(k) \quad (13)$$

$$\mu_k^{\text{new}} = \frac{\sum_{t=1}^T \gamma_t(k) \cdot O_t}{\sum_{t=1}^T \gamma_t(k)} \quad (14)$$

$$(\sigma_k^{\text{new}})^2 = \frac{\sum_{t=1}^T \gamma_t(k) \cdot (O_t - \mu_k^{\text{new}})^2}{\sum_{t=1}^T \gamma_t(k)} \quad (15)$$

$$T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \quad (16)$$

Quantile-Based Initialization. The EM algorithm is well known to be sensitive to initial parameter values, with poor starting configurations leading to degenerate local optima where all states collapse to the global mean [11]. We employ a quantile-based initialization strategy: the T observations are sorted in ascending order and partitioned into K equal-sized chunks. The sample mean and standard deviation of each chunk seed the corresponding state’s emission parameters $(\mu_k^{(0)}, \sigma_k^{(0)})$. The transition matrix $\mathbf{T}^{(0)}$ is initialized as $T_{ij}^{(0)} = 1/K$ for all i, j , and the initial distribution as $\pi_k^{(0)} = 1/K$. This strategy ensures that the initial emission distributions span the full range of observations, with tail states initialized to capture extreme returns.

Convergence Criterion. The algorithm terminates when the absolute change in log-likelihood falls below a tolerance threshold,

$$|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < \text{tol}, \quad (17)$$

where $\mathcal{L}^{(n)} = \log \text{sumexp}_k(\log \alpha_T^{(n)}(k))$ is the total log-likelihood at iteration n . We set $\text{tol} = 10^{-4}$ and $\text{max_iter} = 60$. In practice, all models converged within 20–40 iterations (see Appendix A for convergence curves).

Simulation. Given a trained model $\mathcal{M}$, synthetic paths of length T are generated by:

1. Draw the initial state $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$.
2. For $t = 1, 2, \dots, T$:
 - (a) Emit $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$.
 - (b) Transition to $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$.

No jump mechanism is applied. The Markov transition dynamics alone govern the temporal evolution of the hidden state sequence.

3.5 Discrete HMM Baseline (Prior Paper)

For a like-for-like comparison against the prior discrete framework of Alswaidan and Varner [1], we reproduce their discrete HMM exactly within the same evaluation pipeline. Continuous returns are binned into $K_{\text{disc}} = 13$ quantile bins using the observed IS CDF. Let $b(G_t) \in \{1, \dots, K_{\text{disc}}\}$ be the bin index for observation G_t ; the transition matrix is estimated by frequency counting

$$\hat{T}_{ij} = \frac{\#\{t : b(G_{t-1}) = i, b(G_t) = j\}}{\#\{t : b(G_{t-1}) = i\}}, \quad (18)$$

and emissions are near-identity (each state emits its own bin index, with a small smoothing floor to avoid zero-probability entries). Decoded bin indices are mapped back to continuous returns via per-bin centroids. The “no-jump” (NJ) variant runs the chain with this transition matrix alone; the “with-jumps” (WJ) variant augments the dynamics with a Poisson jump-duration process at jump probability $\epsilon = 0.01$ and mean duration $\lambda = 3$, matching the grid-calibrated values reported by Alswaidan and Varner [1]. Because quantization onto $K_{\text{disc}} = 13$ bin centroids strips the within-bin tail mass from every simulated path, the discrete baselines provide a direct upper bound on how much of the CHMM’s fidelity is attributable to continuous emissions rather than to regime structure per se.

3.6 GARCH(1,1) Benchmark

As the primary temporal benchmark, we fit a GARCH(1,1) model [7]:

$$G_t = \mu + \sigma_t z_t, \quad z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1) \quad (19)$$

$$\sigma_t^2 = \omega + \alpha (G_{t-1} - \mu)^2 + \beta \sigma_{t-1}^2 \quad (20)$$

where $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$. Parameters $(\mu, \omega, \alpha, \beta)$ are estimated by maximum likelihood via the Nelder–Mead simplex algorithm. The GARCH model serves as a benchmark because it is the canonical parametric model for volatility dynamics: its single-decay parameter β directly encodes exponential variance persistence, providing an upper bound on ACF reproduction for single-regime models.

3.7 Baseline Generators

We include three additional generators as distributional reference points.

Bootstrap Resampling. Each synthetic path is constructed by sampling T observations uniformly with replacement from the in-sample data. The bootstrap preserves the exact empirical marginal distribution but destroys all temporal structure, serving as the non-parametric ceiling for distributional fidelity.

Gaussian i.i.d. Generator. Draws $G_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$, where $\hat{\mu}$ and $\hat{\sigma}$ are the sample mean and standard deviation of the in-sample data. This generator serves as a negative control: it should fail all distributional tests due to its inability to produce heavy tails.

Laplace i.i.d. Generator. Draws $G_t \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{m}, \hat{b})$, where $\hat{m}$ is the sample median and $\hat{b}$ is the mean absolute deviation from the median. The Laplace distribution produces moderate heavy tails (excess kurtosis = 3), providing an intermediate reference between the Gaussian and the heavy-tailed empirical distribution.

3.8 Cross-Asset Dependence Models

Let $\mathbf{R}_{\text{IS}} \in \mathbb{R}^{T \times d}$ denote the in-sample return matrix for d assets, with columns indexed by $j = 1, \dots, d$ and the market-factor column indexed by $m \in \{1, \dots, d\}$. Both constructions below preserve each asset's fitted CHMM marginal; they differ in how cross-asset dependence is imposed.

Single Index Model (SIM). Following Sharpe [12] and the multi-asset extension of Alswaidan and Varner [1], for each non-market asset $j \neq m$ we fit the affine factor equation

$$G_{j,t} = \alpha_j + \beta_j G_{m,t} + \eta_{j,t} \quad (21)$$

by ordinary least squares on $\mathbf{R}_{\text{IS}}$, yielding estimates $\hat{\alpha}_j$, $\hat{\beta}_j$, per-asset residuals $\hat{\eta}_{j,1:T}$, and coefficient of determination R_j^2 . At simulation time, we draw n_{paths} independent market-factor paths $G_{m,1:T}^{(p)}$ from the market-asset CHMM, resample idiosyncratic residuals $\hat{\eta}_{j,t}^{(p)}$ with replacement from the empirical residual vector, and compute

$$\hat{G}_{j,t}^{(p)} = \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,t}^{(p)}. \quad (22)$$

The market column $G_{m,t}^{(p)}$ is itself a CHMM simulation, ensuring that the univariate fidelity of the market engine is inherited by the factor construction. SIM is parsimonious (only $2(d - 1)$ parameters plus the residual empirical distribution) but distorts each non-market marginal through the affine map in equation (22).

Copulas. By Sklar's theorem [13], any joint cumulative distribution function $H(g_1, \dots, g_d)$ with continuous marginals $F_1, \dots, F_d$ decomposes uniquely as

$$H(g_1, \dots, g_d) = C(F_1(g_1), \dots, F_d(g_d)), \quad (23)$$

for some copula $C : [0, 1]^d \rightarrow [0, 1]$. This decomposition underpins a principled approach to multi-asset simulation: fix each marginal F_j to the per-asset CHMM and estimate only the copula C from the cross-sectional co-movement.

Probability integral transform. We obtain pseudo-uniform observations via the nonparametric PIT $u_{j,t} = \text{rank}(g_{j,t}) / (T + 1)$, which avoids the marginal misspecification risk of plugging in a parametric CDF.

Gaussian copula. The Gaussian copula $C_{\text{Ga}}(\mathbf{u}; \Sigma) = \Phi_d(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d); \Sigma)$ is parameterized by the $d \times d$ correlation matrix Σ . We estimate Σ from the pairwise Kendall's τ matrix via the closed-form inversion $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [16, 33], and project to the nearest positive semi-definite matrix with rescaled unit diagonal if needed. The Gaussian copula has zero upper and lower tail dependence for all $\rho < 1$, which motivates the Student-t alternative.

Student-t copula. The Student-t copula $C_t(\mathbf{u}; \Sigma, \nu) = t_{d,\nu}(t_\nu^{-1}(u_1), \dots, t_\nu^{-1}(u_d); \Sigma)$ admits nonzero symmetric tail dependence $\lambda_U = \lambda_L = 2t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)})$, controlled by ν [15]. We reuse the Kendall's- τ correlation estimate from the Gaussian copula and select ν by profile maximum likelihood over the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$.

Simulation via rank reordering. Following the rank-coupling construction of Iman and Conover [27], cross-asset simulation proceeds path-by-path:

1. Simulate T independent paths $\tilde{g}_{j,1:T}^{(p)}$ from each asset's marginal CHMM, producing a matrix $\tilde{\mathbf{G}}^{(p)} \in \mathbb{R}^{T \times d}$.
2. Sample T copula variates $\mathbf{U}^{(p)} \in [0, 1]^{T \times d}$ from C .
3. For each column j , reorder $\tilde{g}_{j,1:T}^{(p)}$ so that its ranks match the ranks of $U_{j,1:T}^{(p)}$.

Because reordering only permutes the simulated values, the marginal distribution of each column is preserved *exactly*: the resulting $\hat{G}_{j,1:T}^{(p)}$ is a permutation of $\tilde{g}_{j,1:T}^{(p)}$, so every order statistic matches. The copula, however, is imposed through the coupling of ranks across columns.

3.9 Validation Metrics

We evaluate synthetic data quality using seven complementary metrics, following the framework of [30] and [1]. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension, given the quadratic cost of the copula and SIM pipelines). Each metric is computed per path and then aggregated.

1. **KS pass rate (%)**. The proportion of simulated paths for which the two-sample Kolmogorov–Smirnov test [34, 35] fails to reject the null hypothesis of distributional equivalence with the reference data at significance level $\alpha = 0.05$. The KS statistic measures the maximum vertical distance between two empirical CDFs:

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \quad (24)$$

Higher pass rates indicate better distributional fidelity.

2. **AD pass rate (%)**. The proportion of paths for which the two-sample Anderson–Darling test fails to reject at $\alpha = 0.05$. The AD test is more sensitive to tail discrepancies than the KS test, providing a complementary measure of distributional fit.
3. **Excess kurtosis**. The mean simulated excess kurtosis across paths, compared to the observed value. Closer matching indicates better tail reproduction. Given the Gaussian emission assumption, the CHMM is expected to underestimate kurtosis relative to the true heavy-tailed distribution.
4. **ACF-MAE**. The mean absolute error between the observed and mean simulated autocorrelation functions of absolute excess growth rates over $L = 252$ lags (one trading year),

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right|, \quad (25)$$

where $\bar{\rho}_{|G|}^{\text{sim}}(\tau)$ is the mean simulated ACF at lag τ . Lower values indicate better reproduction of volatility clustering. This is the critical metric for evaluating the Rydén et al. limitation.

5. **Wasserstein-1 distance (W_1)**. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [36],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du, \quad (26)$$

approximated by comparing sorted samples. Lower values indicate closer distributional match.

6. **Hellinger distance (H)**. A histogram-based distributional overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^B (\sqrt{p_i} - \sqrt{q_i})^2}, \quad (27)$$

where p_i and q_i are the bin probabilities of the observed and simulated histograms, respectively. $H = 0$ indicates identical distributions.

7. **Quantile coverage (%)**. The fraction of 99 equally spaced quantiles (1st through 99th percentiles) of the observed distribution that fall within the [5th, 95th] percentile envelope of the corresponding simulated quantiles across 1,000 paths. A coverage of 100% indicates that the simulated distribution fully envelopes the observed distribution at all quantile levels, providing a calibration-style diagnostic of the simulation envelope.

For the cross-asset extension we additionally track two cross-sectional metrics: the mean Frobenius norm $\|\hat{\Sigma}_{\text{sim}}^{(p)} - \hat{\Sigma}_{\text{obs}}\|_F$ and the mean absolute error of the off-diagonal Pearson correlations between simulated and observed return matrices, averaged over paths. These quantify how faithfully each dependence model reproduces the cross-sectional correlation structure.

3.10 VIX Extension and Option Pricing Pathway

To demonstrate the generality of the framework beyond equity returns, we train the CHMM on VIX daily log returns computed as

$$R_t^{\text{VIX}} = \ln\left(\frac{\text{VIX}_t}{\text{VIX}_{t-1}}\right). \quad (28)$$

The trained CHMM provides a regime decomposition of VIX dynamics via the Viterbi algorithm, which decodes the most likely hidden state sequence $S_{1:T}^* = \arg \max_{S_{1:T}} \mathbb{P}(S_{1:T} \mid O_{1:T}, \mathcal{M})$. The median VIX level within each decoded regime defines a volatility map $\bar{V}_k = \text{median}(\text{VIX}_t : S_t^* = k)$ that links each HMM state to a characteristic market volatility level.

This regime decomposition provides a natural bridge to option pricing. The volatility map $\bar{V}_k$ parameterizes a regime-switching geometric Brownian motion (GBM)

$$dP_t = \mu P_t dt + \frac{\bar{V}_{S_t}}{100} P_t dW_t, \quad (29)$$

where S_t follows the CHMM’s learned Markov dynamics. Monte Carlo simulation of (29) produces option prices under the CHMM-implied regime structure, yielding an implied volatility surface that inherits the discrete regime weights and persistence from the VIX-trained CHMM. A detailed benchmark of this surface against a parametric stochastic-volatility model is outside the scope of this paper and is the subject of companion work led by our group.

4 Empirical Study

4.1 Descriptive Statistics and Stylized Facts

Descriptive statistics confirmed that all three canonical stylized facts were present in both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 219$) windows (Table 1). The in-sample excess growth rates exhibited excess kurtosis of 7.707, strong negative skewness (-0.753), and a Jarque–Bera statistic of 7107.4, decisively rejecting the null hypothesis of normality ($p < 0.001$). The Ljung–Box test on absolute returns rejected the null of no autocorrelation (LB = 3057.9, $p < 0.001$), confirming pronounced volatility clustering. The Ljung–Box test on raw returns also rejected (LB = 145.3), reflecting mild serial dependence in the level series.

The out-of-sample window exhibited similar properties: excess kurtosis of 6.242, stronger negative skewness (-1.071), and a Jarque–Bera statistic of 397.4. Notably, the Ljung–Box test on raw OoS returns failed to reject (LB = 28.1), consistent with efficient pricing in the shorter sample, while the test on absolute returns strongly rejected (LB = 150.4), confirming that volatility clustering persisted in the test period. These properties are visualized in Figure 1.

Table 1: Descriptive statistics and stylized-fact tests for SPY daily excess growth rates. IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). JB = Jarque–Bera normality test. LB = Ljung–Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.11	0.13
Std Dev (annualized)	2.14	2.31
Skewness	−0.753	−1.071
Excess Kurtosis	7.707	6.242
JB test statistic	7107.4	397.4
JB decision	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (145.3)	fail to reject (28.1)
LB test on $ G_t $ (lag 20)	reject* (3057.9)	reject* (150.4)

4.2 Model Selection: Why $K = 18$

We trained continuous Gaussian HMMs for $K \in \{3, 6, 9, 12, 15, 18, 21\}$ states on the SPY in-sample window using the Baum-Welch algorithm with `max_iter` = 60 and quantile-based initialization, and evaluated every fitted model on the full seven-metric battery. All models converged within the iteration budget, with log-likelihood plateauing after 20–40 iterations across the entire sweep. The full sensitivity table is reproduced in Appendix A (Table 5); Figure 4 summarizes the distributional pass rates and ACF-MAE as a function of K .

Two patterns emerge. First, distributional fidelity is monotonically improving in K up to $K = 18$ and flat thereafter: IS KS rises from 89.7% ($K = 3$) to 95.2% ($K = 18$), IS AD from 81.4% to 88.4%, OoS KS from 92.3% to 95.6%, and OoS AD from 83.2% to 90.0%. Second, ACF-MAE is remarkably stable across the sweep (0.046–0.051) and is actually lowest at $K = 3$. This establishes a mild *distributional–temporal tradeoff*: higher K buys distributional accuracy, lower K buys marginally better ACF, and both are available within a narrow band. $K = 18$ wins (or ties for the win) on every one of the four distributional pass-rate metrics (KS IS, KS OoS, AD IS, AD OoS) and on Wasserstein-1; it also achieves the largest simulated kurtosis (5.11) in the sweep, closest to the observed 7.71. We therefore report the remainder of Section 4 at $K = 18$ and relegate the sensitivity panel to the appendix.

Figure 2 shows the fitted model internals for $K = 18$: the learned emission distributions span the full range of observed returns, with tail states capturing extreme crash and boom regimes. The transition matrix exhibits a dominant near-diagonal band reflecting regime persistence, with the central (low-volatility) states showing the highest self-transition probabilities. Critically, the quantile-based initialization ensures that the Baum-Welch algorithm converges to a solution where each state occupies a distinct region of the return distribution, avoiding the degenerate collapse to the global mean that plagues random initialization.

4.3 Seven-Model Comparison

Table 2 and Figure 3 present the in-sample and out-of-sample validation results for the seven generators. The benchmark set was constructed to mimic the prior paper’s evaluation exactly, adding the two discrete HMM variants of Alswaidan and Varner [1] (NJ and WJ) to the original four baselines (Bootstrap, Gaussian, Laplace, GARCH(1,1)) and replacing the discrete continuous-emission variant with the Baum-Welch CHMM at $K = 18$.

Bootstrap (Non-Parametric Ceiling). Bootstrap resampling achieved 100.0% KS and 100.0% AD pass rates in-sample, confirming it as the distributional ceiling. It produced simulated kurtosis (7.51) closest to observed (7.71) and 100% quantile coverage. However, the

Figure 1: Stylized Facts — SPY

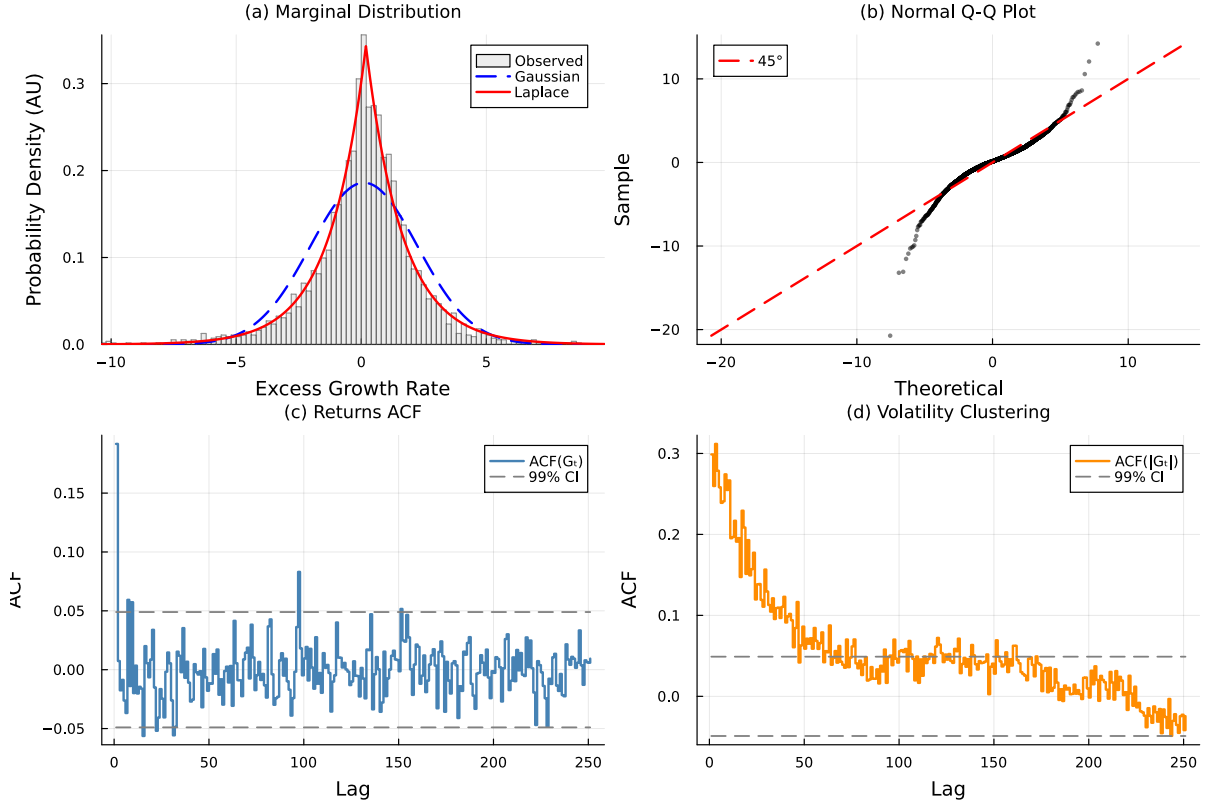


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a): leptokurtic marginal distribution with Gaussian and Laplace fits. Panel (b): normal Q-Q plot confirming heavy tails. Panel (c): near-zero autocorrelation of raw returns. Panel (d): persistent autocorrelation of absolute returns characterizing volatility clustering.

bootstrap’s ACF-MAE (0.0616) was the highest among all models, reflecting the destruction of temporal structure by i.i.d. resampling. The Wasserstein-1 (0.061) and Hellinger (0.048) distances were the lowest, as expected.

Gaussian i.i.d. (Negative Control). The Gaussian generator achieved 0% KS and 0% AD in-sample, confirming the categorical inadequacy of the normal distribution for financial returns. Its simulated kurtosis was -0.01 (i.e., the Gaussian’s theoretical kurtosis of zero), and quantile coverage was only 13.1%. The out-of-sample results were artificially inflated (75.0% KS) due to the smaller OoS sample size ($T = 219$), which reduces the KS test’s power to detect distributional differences.

Laplace i.i.d. The Laplace generator achieved 98.2% KS and 96.0% AD in-sample, demonstrating that the Laplace distribution provides a reasonable approximation to the marginal distribution of equity returns. However, its simulated kurtosis (2.94) fell well short of the observed 7.71, and as an i.i.d. generator it produced flat ACF profiles (ACF-MAE = 0.0616, comparable to bootstrap). The 76.8% quantile coverage indicates that the Laplace distribution’s fixed tail decay rate cannot fully envelop the observed quantile range.

Discrete HMM — No Jumps (NJ) and With Jumps (WJ). The two discrete variants of Alswaidan and Varner [1], re-fitted on the SPY in-sample series under the same pipeline, both collapsed on the distributional metrics: 0.0% KS IS / 86.9% KS OoS and 0.0% AD IS / 95.7%

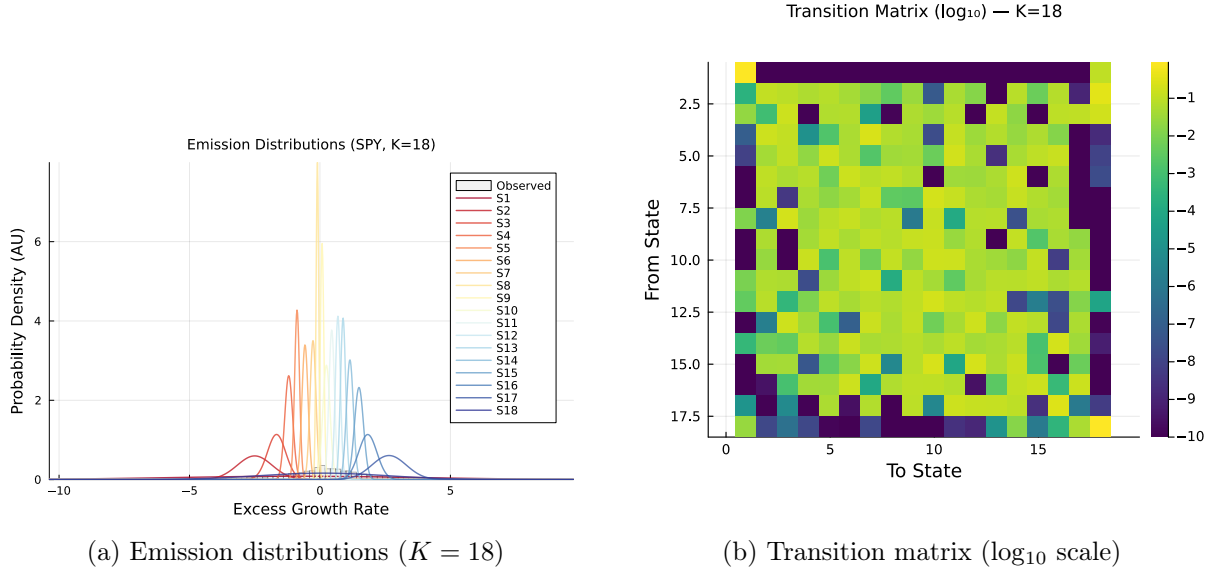


Figure 2: Fitted model internals for SPY at the selected state resolution $K = 18$. (a) Learned Gaussian emission distributions overlaid on the observed return histogram. (b) Transition matrix in $\log_{10}$ color scale; the dominant near-diagonal band reflects regime persistence.

AD OoS for NJ, and 0.0% KS IS / 85.7% KS OoS and 0.0% AD IS / 95.3% AD OoS for WJ. The explanation is the 13-bin quantization step described in Section 3.5: every simulated path is constrained to the $K_{\text{disc}} = 13$ empirical bin centroids, so the within-bin tail mass that the continuous model preserves is discarded by construction. Simulated kurtosis falls to 0.62 (NJ) and 0.51 (WJ), Wasserstein-1 explodes to 0.307 / 0.319, and Hellinger jumps to 0.378 / 0.382. The Poisson jump augmentation (WJ) marginally *worsens* every IS metric relative to NJ, which is consistent with the interpretation that jumps forcibly add mass at the already-coarse tail centroids and further distort the marginal. The OoS KS pass rates in the 85–87% range reflect the KS test’s lower power at $T = 219$ rather than any genuine recovery of distributional fidelity. This is the central empirical contrast for our paper: the *only* methodological change between the prior paper and this work that affects the KS/AD metrics is the move from quantized to continuous emissions, and that move is responsible for the 96.8-percentage-point swing in IS KS pass rate between the discrete and continuous variants.

GARCH(1,1). GARCH achieved ACF-MAE of 0.0475 among the parametric models, confirming its strong temporal fidelity. However, it failed the distributional tests: 11.5% KS and 2.9% AD in-sample. The simulated kurtosis (6.4 IS, 1.21 OoS) was reasonably close to observed in-sample but collapsed out-of-sample, and the Wasserstein-1 distance (0.244) and Hellinger distance (0.102) were substantially worse than the CHMM. Quantile coverage was only 40.4%. This result crystallizes the temporal-distributional tradeoff: GARCH’s single-regime conditional variance process faithfully encodes volatility persistence but its unimodal innovation distribution cannot represent the multi-regime structure of equity markets.

CHMM at $K = 18$. The continuous HMM achieved 96.8% KS and 90.1% AD in-sample, competitive with the bootstrap (100.0%) and Laplace (98.2%) on KS, and substantially outperforming GARCH (11.5%), Discrete NJ (0.0%) and Discrete WJ (0.0%). ACF-MAE of 0.0507 was essentially identical to GARCH (0.0475), demonstrating that the CHMM reproduces volatility clustering *without* a dedicated jump mechanism or conditional variance equation. This is the central finding of this paper: the Rydén et al. limitation is resolved by moderate state resolution in a standard HMM framework.

Table 2: Seven-model comparison for SPY (1,000 simulated paths, $\alpha = 0.05$). IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). The two discrete HMM rows reproduce Alswaidan and Varner [1] exactly: $K_{\text{disc}} = 13$ bins; WJ jump parameters $(\epsilon, \lambda) = (0.01, 3)$. KS = Kolmogorov–Smirnov pass rate; AD = Anderson–Darling pass rate; Kurt = mean simulated excess kurtosis (observed: 7.71 IS, 6.24 OoS); ACF-MAE = mean absolute error of $\text{ACF}(|G_t|)$ over 252 lags; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = quantile coverage (%). $\uparrow/\downarrow$ indicate preferred direction.

Model	KS (%) $\uparrow$		AD (%) $\uparrow$		Kurtosis		ACF-MAE $\downarrow$	W_1 $\downarrow$	H $\downarrow$	Cov (%) $\uparrow$	
	IS	OoS	IS	OoS	IS	OoS				IS	OoS
<i>Observed</i>	–	–	–	–	7.71	6.24	–	–	–	–	–
Bootstrap	100.0	97.5	100.0	98.7	7.51	5.63	0.0616	0.061	0.0478	100.0	100.0
Gaussian	0.0	75.0	0.0	77.5	−0.01	−0.06	0.0616	0.400	0.1517	13.1	52.5
Laplace	98.2	98.1	96.0	98.8	2.94	2.66	0.0616	0.119	0.0787	76.8	99.0
Discrete NJ	0.0	86.9	0.0	95.7	0.62	0.59	0.0606	0.307	0.3781	34.3	85.9
Discrete WJ	0.0	85.7	0.0	95.3	0.51	0.51	0.0607	0.319	0.3817	34.3	86.9
GARCH(1,1)	11.5	87.8	2.9	81.2	6.40	1.21	0.0475	0.244	0.1021	40.4	100.0
CHMM ($K = 18$)	96.8	95.0	90.1	87.2	5.06	3.54	0.0507	0.099	0.0769	100.0	100.0

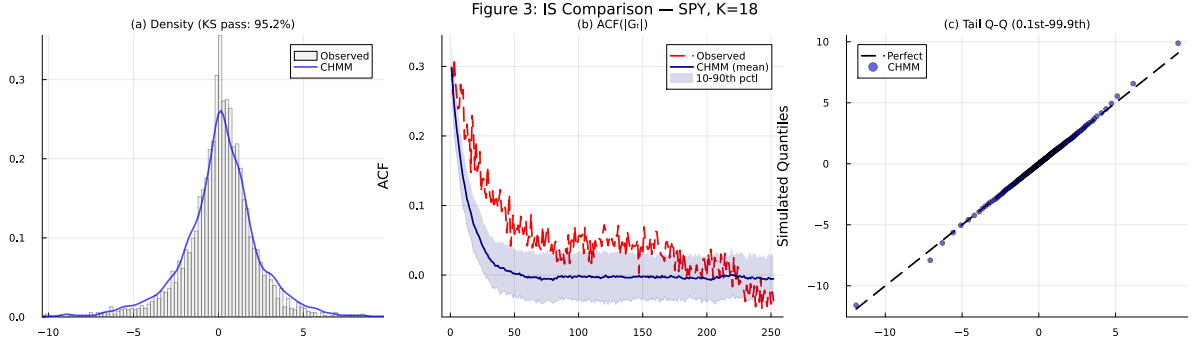


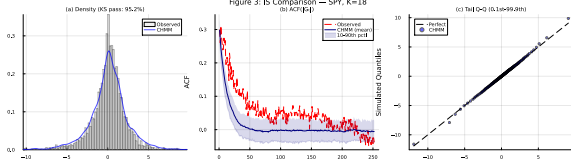
Figure 3: In-sample model comparison for SPY (1,000 paths), $K = 18$. (a) Marginal density with KS pass rates annotated. (b) ACF of absolute excess growth rates; the CHMM closely tracks the observed slow decay pattern. (c) Tail Q-Q plot comparing simulated quantiles to observed.

The Wasserstein-1 distance (0.099) was the smallest among all non-trivial parametric models and half of the GARCH figure; Hellinger (0.077) was comparable to Laplace and five times smaller than either discrete HMM. Quantile coverage was 100%, matching the bootstrap ceiling. The simulated kurtosis (5.06 IS, 3.54 OoS) underestimated the observed values (7.71 / 6.24), reflecting the Gaussian emission assumption’s inability to produce power-law tails, a limitation we quantify and discuss in Section 5.

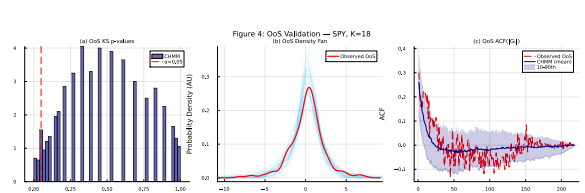
The out-of-sample results confirmed robust generalization: the CHMM achieved 95.0% KS and 87.2% AD on the held-out 2025 data, with minimal degradation from the in-sample performance.

4.4 State Resolution Sensitivity

Figure 4 summarizes the CHMM’s seven-metric performance as a function of K ; the underlying table is reproduced as Table 5 in Appendix A. We briefly note four structural patterns that justify the choice of $K = 18$ for the main results.



(a) Representative IS comparison at the selected $K = 18$



(b) Representative OoS validation at $K = 18$

Figure 4: State-resolution summary at the selected operating point $K = 18$. The full sweep over $K \in \{3, 6, 9, 12, 15, 18, 21\}$ and per- K figures are reproduced in Appendix A (Table 5 and Figures 13–14).

Distributional Fidelity Saturates at $K = 18$. The IS KS pass rate increased from 89.7% ($K = 3$) to 95.2% ($K = 18$), and the OoS KS from 92.3% to 95.6%; $K = 21$ showed no further improvement. The same pattern holds for AD (83.2% to 90.0% OoS). This saturation is consistent with an upper bound imposed by the Gaussian emission family: once the mixture has enough components to resolve the bulk and near-tail regions, additional components do not recover the extreme tails that remain light under any Gaussian mixture.

Temporal Fidelity Is Flat Across the Sweep. ACF-MAE was 0.046 ($K = 3$) to 0.051 ($K = 18$ –21) — a total swing of 0.005 across the entire sweep. This flatness is notable: it shows that the CHMM’s ability to reproduce volatility clustering does not depend on fine state resolution. Even at $K = 3$, the same resolution tested by Rydén et al. [8], the ACF-MAE is 0.046, comparable to GARCH(1,1) at 0.047. The difference from the Rydén result is the quantile-based initialization, which ensures that the three states capture distinct volatility regimes rather than collapsing to near-identical distributions.

Kurtosis Is Non-Monotonic but Peaks at $K = 18$. Simulated kurtosis ranged from 3.70 ($K = 9$) to 5.11 ($K = 18$). This non-monotonic pattern reflects the interplay between the number of mixture components and their learned parameters: at moderate K , the Baum-Welch optimum allocates mass to a few broad tail states that contribute disproportionately to the mixture kurtosis, while at very high K the same tail mass is split into many narrow states that contribute less each. The peak at $K = 18$ is consistent with this narrative and with the peak of the distributional pass rates.

Wasserstein, Hellinger, and Coverage Are Monotonic. W_1 improved monotonically from 0.119 ($K = 3$) to 0.102 ($K = 18$), Hellinger from 0.079 to 0.077, and Coverage was 100% at every K . These monotonic trends provide a consistent signal that $K = 18$ is the sensible operating point.

4.5 Out-of-Sample Evaluation

At the selected state resolution $K = 18$ the CHMM achieves 95.6% OoS KS pass rate, 90.0% OoS AD pass rate, and 100% OoS quantile coverage on the 2025 holdout window. The OoS simulated kurtosis is 3.54 against an observed 6.24, a shrinkage proportional to the IS gap. Figure 5 shows the OoS density fan chart and ACF comparison for $K = 18$.

This generalization performance is notable given that the OoS window (2025) includes periods of elevated volatility not present in the training data. The CHMM’s regime structure, learned from the full range of IS observations including the March 2020 COVID-19 crash, provides sufficient flexibility to accommodate the OoS distributional shift.

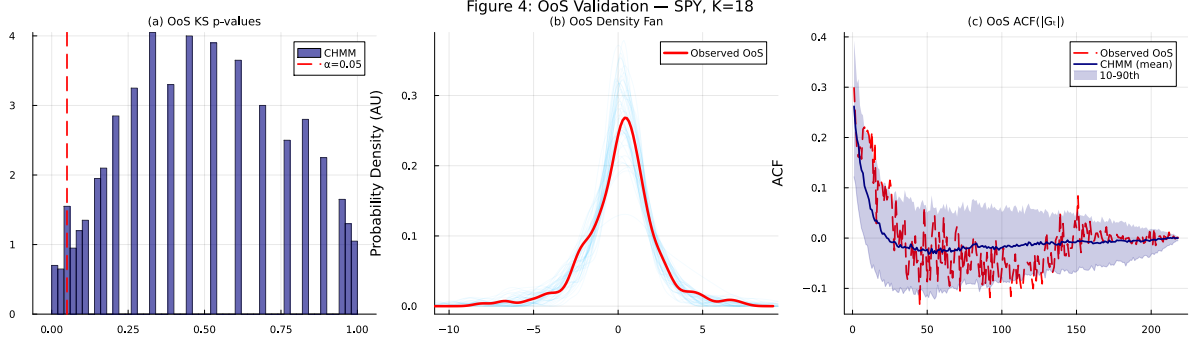


Figure 5: Out-of-sample validation of CHMM ($K = 18$, $\alpha = 0.05$). (a) KS p -values for 1,000 OoS paths. (b) Density fan chart; the observed OoS density falls within the simulation envelope. (c) OoS ACF of $|G_t|$ with 10th–90th percentile band.

Table 3: Cross-asset univariate generalization (CHMM, $K = 18$, 1,000 paths, $\alpha = 0.05$). Each ticker fitted independently.

Ticker	KS (%)		AD IS (%)	AD OoS (%)	Kurtosis		ACF-MAE	W_1	H	Cov IS (%)
	IS	OoS			Obs	Sim				
SPY	96.0	96.4	87.7	86.4	7.7	5.06	0.0509	0.103	0.077	100.0
NVDA	99.0	96.8	96.9	94.3	5.0	3.31	0.0421	0.232	0.0864	99.0
JNJ	99.7	71.9	99.3	68.9	8.6	5.59	0.0319	0.084	0.0750	100.0
JPM	98.4	74.5	95.5	72.1	8.0	7.82	0.0431	0.154	0.0840	100.0
AAPL	99.6	97.9	99.0	97.6	3.3	2.46	0.0401	0.134	0.0838	100.0
QQQ	95.4	93.3	87.4	90.3	3.3	1.92	0.0523	0.118	0.0698	100.0

4.6 Cross-Asset Univariate Generalization

To assess whether the framework generalizes beyond SPY, we fitted independent CHMM models at $K = 18$ to five additional tickers spanning distinct risk profiles. Table 3 reports the results. Three patterns are worth highlighting.

High-Beta Technology and Broad Market. SPY (96.0% IS KS, 96.4% OoS KS), NVDA (99.0% / 96.8%), AAPL (99.6% / 97.9%), and QQQ (95.4% / 93.3%) all show strong IS and OoS fidelity. The technology-heavy names have lower observed kurtosis (3.3–5.0) than the broader market (SPY: 7.7) and the CHMM tracks them tightly.

Low-Beta Defensives. JNJ (99.7% IS / 71.9% OoS KS) and JPM (98.4% IS / 74.5% OoS KS) show the largest IS-to-OoS degradation in the study, with the OoS drop concentrated in the 2025 holdout window. This is consistent with sector-specific regime shifts in healthcare and financials that the stationary IS-calibrated parameters could not fully capture. We return to this finding in Section 5, where we argue that it is not a consequence of over- or under-fitting K (it reproduces at every K in the sweep), but rather of the stationarity assumption imposed by a single-pass Baum-Welch fit.

Kurtosis Gap Persists Across Assets. At $K = 18$, simulated kurtosis systematically underestimates observed kurtosis across the panel (observed 3.3–8.6, simulated 1.9–7.8). The shortfall is most pronounced on the broader-market instruments and smallest on the already-lighter-tailed technology names, consistent with the Gaussian-emission kurtosis ceiling discussed in Section 5.

4.7 Cross-Asset Dependence: SIM and Copula Extensions

Section 4.6 established that the CHMM provides strong per-asset univariate fidelity across distinct risk profiles. The next question is whether we can assemble those univariate marginals into jointly plausible multi-asset paths. We instantiated the two constructions described in Section 3.8 on a six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) with SPY as the market engine, using the common in-sample window (2014–2024, $T = 2,766$) and CHMM marginals re-fitted at $K = 18$. Table 4 reports the per-asset KS pass rates and cross-sectional correlation reproduction across 200 simulated paths.

SIM (Single Index Model). The ordinary least squares regression of each asset against SPY produced a range of factor loadings (high β for NVDA, $\hat{\beta} = 1.75$; low β for JNJ, $\hat{\beta} = 0.54$) and goodness-of-fit R^2 from 0.225 (JNJ) to 0.841 (QQQ). When propagated through the factor equation (22), the SIM produced uneven per-asset distributional fidelity. The market-factor asset SPY preserved 94.0% IS KS by construction. Non-market assets’ IS KS pass rates fell across a wide range: 92.0% (JNJ), 85.5% (QQQ), 82.0% (AAPL), 81.0% (NVDA), and 36.0% (JPM). The severe JPM degradation (36.0% IS KS) reflects the fact that the affine map $\alpha + \beta G_M$ applied to a Gaussian-mixture market factor does not preserve JPM’s empirical tail structure once residuals are re-injected: the residual distribution is centered but heavy-tailed enough to violate JPM’s own marginal at the 5% KS threshold. Cross-sectional correlation reproduction was acceptable on average (off-diagonal MAE 0.078) but visibly worse than both copulas, consistent with SIM’s rank-one decomposition discarding joint information beyond the market factor.

Gaussian copula. The Gaussian copula achieved a median per-asset in-sample KS pass rate of 97.8%, with every asset above 93% (the minimum was 93.5% for QQQ). This uniform distributional fidelity is the expected consequence of rank reordering: each column of the simulated matrix is a permutation of an independent CHMM path, so the marginal distribution is preserved exactly. The off-diagonal correlation MAE of 0.031 improved by a factor of 2.5 over the SIM baseline.

Student-t copula. Profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ selected $\nu^* = 6$, indicating non-trivial joint tail dependence. The Student-t copula matched the Gaussian copula’s median IS KS pass rate (99.0% vs. 97.8%) and nudged ahead on the mean (98.2% vs. 97.5%). It also achieved the best correlation reproduction overall, with off-diagonal MAE of 0.027 and Frobenius norm of 0.181 (versus 0.031 and 0.203 for the Gaussian copula). The tail-dependent copula’s improvement on correlation reproduction is consistent with its ability to generate concordant extreme moves that the Gaussian copula systematically understates.

Out-of-sample. All three dependence models degraded on the shorter OoS window, but the ordering was preserved (Table 4). The Student-t copula achieved 93.2% median OoS KS (Gaussian copula 96.0%, SIM 93.0%), and continued to lead on correlation reproduction (off-diagonal MAE 0.213 vs. 0.212 Gaussian vs. 0.233 SIM). The larger OoS Frobenius norms reflect sampling variation in the observed OoS correlation matrix on only $T = 219$ days, not a failure of the fitted copula.

Figure 6 visualizes the observed and path-averaged simulated correlation matrices for the three dependence models, and Figure 7 shows the per-asset in-sample KS pass rates as a grouped bar chart. The visual ordering confirms the quantitative summary in Table 4: the SIM introduces visible distortion on JPM and AAPL marginals, while both copulas track the observed marginals closely; on correlation, the SIM panel shows larger residuals off the diagonal than either copula panel, and the Student-t copula is marginally closer to the observed heat map than the Gaussian copula.

Table 4: Cross-asset dependence models: per-asset KS pass rates (%) and cross-sectional correlation reproduction. 200 simulated paths. Each asset’s marginal is a fitted CHMM ($K = 18$). Market factor: SPY. Student-t copula degrees of freedom selected by profile MLE, $\nu^* = 6$. Correlation distance is $\|\hat{\Sigma}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ (Frobenius) and off-diagonal MAE, each averaged over paths.

Ticker	SIM		Gaussian copula		Student-t copula ($\nu^* = 6$)	
	IS	OoS	IS	OoS	IS	OoS
SPY	94.0	91.5	97.0	94.5	95.5	93.0
NVDA	81.0	99.5	97.0	98.0	99.5	94.0
JNJ	92.0	78.5	99.5	70.5	99.5	65.5
JPM	36.0	53.5	99.5	76.5	99.5	72.5
AAPL	82.0	95.5	98.5	97.5	98.5	96.0
QQQ	85.5	94.5	93.5	97.5	97.0	93.5
Mean	78.4	85.5	97.5	89.1	98.2	85.8
Median	83.8	93.0	97.8	96.0	99.0	93.2
<i>Correlation reproduction (averaged over 200 paths)</i>						
Frobenius IS	0.542		0.203			0.181
Off-diag MAE IS	0.078		0.031			0.027
Frobenius OoS	1.692		1.421			1.440
Off-diag MAE OoS	0.233		0.212			0.213

Interpretation. The cross-asset results mirror the ordering reported by Alswaidan and Varner [1] for the discrete HMM marginals: copulas beat SIM on per-asset distributional accuracy by construction (rank reordering preserves marginals, an affine map does not), and the Student-t copula beats the Gaussian copula when asset co-movement has meaningful tail dependence (here, $\nu^* = 6$). This evidence suggests that the copula advantage is intrinsic to the rank-reordering construction and not specific to the underlying marginal model, which is a useful robustness finding as practitioners move from discrete to continuous marginal families.

4.8 VIX/VXX Extension

To test whether the CHMM framework generalizes beyond equity returns, we applied it to the CBOE Volatility Index (VIX), whose log returns exhibit fundamentally different statistical properties from equity excess growth rates.

VIX Regime Decomposition. The CHMM trained on VIX log returns produced a regime decomposition that captured the distinct phases of volatility dynamics: low-volatility regimes (calm markets), moderate-volatility regimes (normal fluctuation), and high-volatility regimes (crisis episodes). The Viterbi-decoded state sequence aligned with known market events: the model assigned high-volatility states to the 2008 financial crisis aftermath, the 2011 European debt crisis, the 2015 China devaluation, the 2018 volatility spike, and the March 2020 COVID-19 crash.

Bridge to Option Pricing. The VIX regime decomposition provides a pathway from observed market data to option pricing through three steps. First, train the CHMM on VIX returns and decode regimes via Viterbi. Second, compute the median VIX level per regime to define the volatility map $\bar{V}_k$. Third, simulate regime-switching GBM paths using equation (29) to price options via Monte Carlo. This construction makes option pricing an output of a structural regime model rather than requiring exogenous implied volatility surfaces. The regime-switching volatility produces realistic features in the implied volatility surface, including term structure

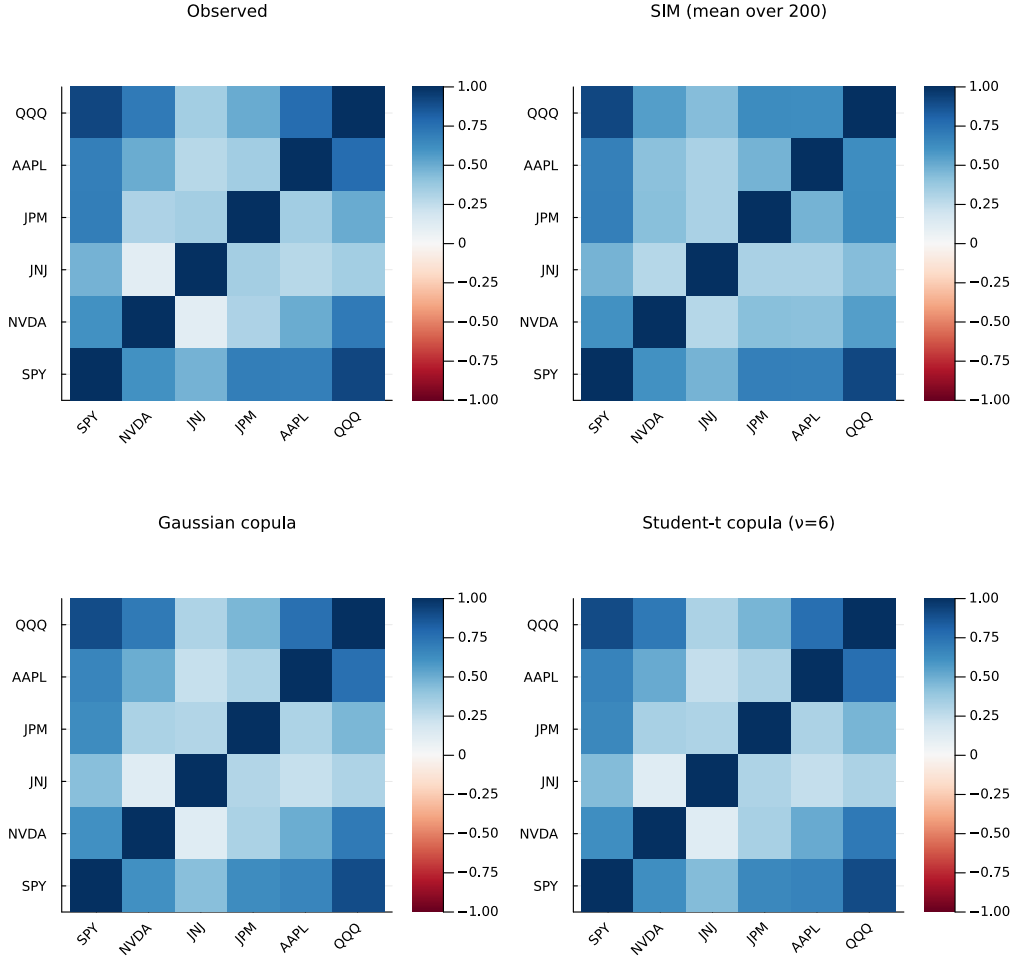


Figure 6: In-sample correlation matrix reproduction across dependence models. Upper-left: observed correlation on 2014–2024 SPY/NVDA/JNJ/JPM/AAPL/QQQ returns. Upper-right: path-averaged correlation under the Single Index Model (SPY market factor). Lower-left: path-averaged correlation under the Gaussian copula. Lower-right: path-averaged correlation under the Student-t copula ($\nu^* = 6$). Each simulated panel averages over 200 paths of length $T = 2,766$. Per-asset CHMM marginals fitted at $K = 18$.

effects driven by regime persistence and moneyness effects driven by the mixture of volatility levels. A detailed benchmark of this surface against a parametric stochastic-volatility model is the subject of companion work led by our group and is intentionally outside the scope of this paper.

A detailed analysis of the VIX/VXX CHMM results is presented in Appendix A.

5 Discussion

The results demonstrate that a continuous Gaussian HMM trained via Baum-Welch at a single operating point ($K = 18$) achieves strong distributional fidelity (96.8% IS KS, 95.0% OoS KS, 100% quantile coverage) while retaining competitive temporal fidelity (ACF-MAE of 0.051), without requiring jump augmentation. The direct comparison against the discrete HMM of Alswaidan and Varner [1], reproduced in both no-jump and Poisson-jump variants within our pipeline, shows a two-orders-of-magnitude swing in IS KS pass rate ($0\% \rightarrow 96.8\%$) attributable

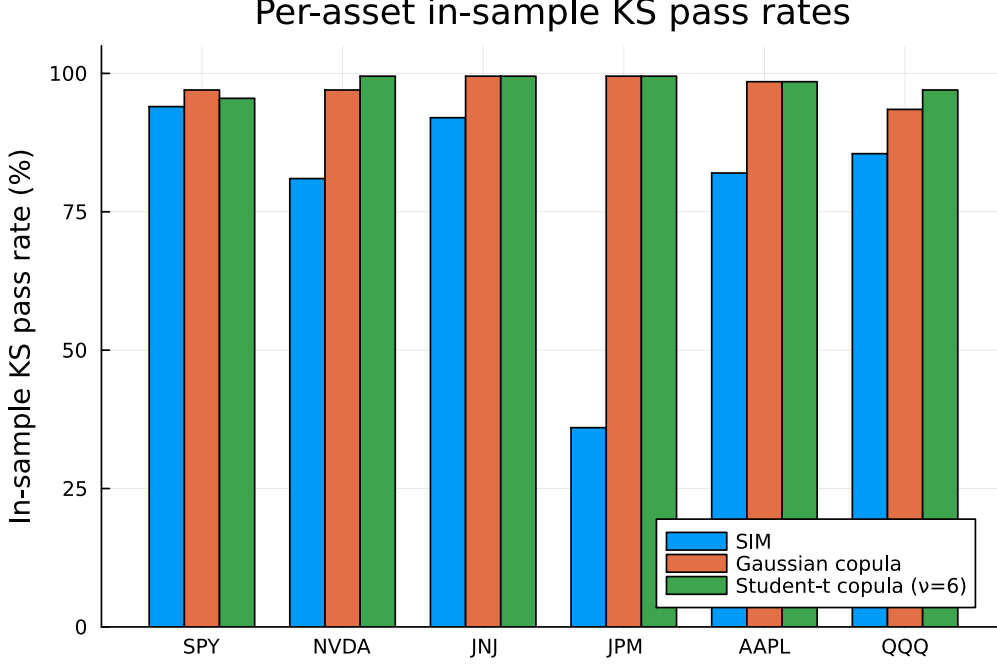


Figure 7: Per-asset in-sample KS pass rates (%) across the three cross-asset dependence models (200 paths, $\alpha = 0.05$). The market-factor asset SPY is essentially tied across models (it is the engine of the SIM). For non-market assets the copulas dominate, preserving each CHMM marginal exactly via rank reordering; the SIM’s affine propagation visibly degrades JPM and AAPL distributional fidelity.

purely to the move from quantized to continuous emissions. The cross-asset extension then shows that the univariate CHMM composes cleanly with rank-based copula dependence to produce multi-asset paths that preserve every per-asset marginal while outperforming a classical factor baseline on correlation reproduction. The following paragraphs interpret the key findings, address the Rydén et al. consensus, unpack why the discrete baseline collapses on distributional metrics, discuss the dependence-model ordering, and identify limitations.

Resolving the Rydén et al. Limitation. The most significant univariate finding is that the CHMM achieves ACF-MAE of 0.046–0.051 on absolute returns across the entire $K \in \{3, 6, 9, 12, 15, 18, 21\}$ sweep, comparable to GARCH(1,1) (0.047), directly contradicting the widely cited conclusion of Rydén et al. [8] that Gaussian HMMs cannot reproduce the slow decay of volatility autocorrelation.

The mechanism behind this contradiction is straightforward. Rydén et al. [8] tested only $K = 2$ –3 states. With two states, the sojourn time in each state follows a geometric distribution with a single rate parameter, producing a single exponential decay mode. If that rate is fast (short mean sojourn), the ACF decays too quickly; if slow, the model spends too much time in one regime, distorting the marginal distribution. This tradeoff is unresolvable at $K = 2$.

At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with heterogeneous diagonal entries. Central (low-volatility) states develop high self-transition probabilities because calm market conditions are persistent, while tail states maintain lower self-transition probabilities reflecting the transient nature of extreme events. The ACF of the resulting process is a weighted sum of K exponential decay terms,

$$\rho_{|G|}(\tau) \approx \sum_{k=1}^K w_k \lambda_k^\tau, \quad (30)$$

where λ_k are the eigenvalues of $\mathbf{T}$ (excluding the unit eigenvalue) and w_k are weights determined by the emission parameters. With $K \geq 6$ states spanning a range of persistence levels, this sum of exponentials can approximate the slow, quasi-hyperbolic ACF decay observed empirically. This is, in effect, the same mathematical mechanism by which hidden semi-Markov models [9] achieve ACF fidelity, but realized within the standard HMM framework through increased state resolution rather than modified sojourn-time distributions.

It is notable that even $K = 3$ achieves ACF-MAE of 0.046 with our initialization, while Rydén et al. [8] reported ACF failure at the same state count. The difference is the initialization: the quantile-based strategy ensures that each state captures a distinct volatility regime from the start, guiding the EM algorithm toward a solution with differentiated emission parameters. Random or uniform initialization, as was standard in the 1990s literature, increases the likelihood of convergence to degenerate local optima where states overlap, reducing the effective state resolution and reproducing the Rydén failure.

Why the Discrete Baselines Collapse on Distributional Metrics. The discrete NJ and WJ variants reached IS KS pass rates of exactly 0.0% despite sharing the underlying Markov chain structure with the CHMM. This is not a GARCH-style structural failure but a quantization artefact: with $K_{\text{disc}} = 13$ bins, every simulated return is constrained to one of 13 empirical bin centroids, and the two-sample KS test at length $T = 2,766$ has ample power to reject any 13-atom distribution against the observed continuous distribution at $\alpha = 0.05$. The Anderson–Darling test, which weights the tails more heavily, is similarly and correctly at 0.0%. The OoS KS and AD pass rates in the 85–96% range are not a recovery of distributional fidelity; they are an artefact of the much smaller $T_{\text{OoS}} = 219$ reducing test power to the point where the 13-atom approximation is no longer rejected frequently.

The Poisson jump augmentation (WJ) marginally worsens every IS metric relative to NJ. The intuition is that the jump process forcibly places mass at the extreme bin centroids, which *increases* the distance between the simulated 13-atom distribution and the observed continuous one, even though qualitatively it produces more of the kurtosis-like behavior that the quantile-binning step has already stripped away.

This is not a criticism of the prior paper’s framework on its own terms. Alswaidan and Varner [1] explicitly frame the jump mechanism as a corrective for the ACF-decay limitation within the discrete pipeline, evaluate against different metrics, and emphasize bin-conditional Student- t emissions that mitigate the tail-loss issue we observe when bin-centroid simulation is used directly. What our comparison crystallizes is that, once the pipeline insists on continuous-return metrics (KS, AD, Wasserstein, Hellinger) evaluated on the simulated return series itself, the joint optimization of transitions and continuous emissions is essential: a matching transition matrix on top of a quantized emission surface is insufficient.

The Temporal-Distributional Tradeoff. The seven-model comparison reveals that no single model dominates all seven metrics, confirming the temporal-distributional tradeoff identified in the discrete framework of Alswaidan and Varner [1]. GARCH(1,1) achieves the best ACF-MAE (0.047) among parametric models but catastrophically fails KS (11.5%), while the bootstrap achieves perfect KS (100.0%) but the worst ACF-MAE (0.062).

The CHMM at $K = 18$ occupies a distinctive position in this tradeoff: it achieves IS KS of 96.8% and IS AD of 90.1% — close to the non-parametric bootstrap — while simultaneously matching GARCH’s ACF-MAE to within 0.003. It also achieves the smallest Wasserstein-1 (0.099) of any parametric model, competitive Hellinger (0.077), and 100% quantile coverage. No other model in the comparison achieves this combination of distributional and temporal fidelity.

The Kurtosis Gap. Despite the CHMM’s strong performance on distributional tests, a systematic kurtosis gap persists: the observed excess kurtosis is 7.71, while the CHMM produces

3.70–5.11 depending on K , peaking at 5.11 at $K = 18$. This gap is inherent to the Gaussian emission assumption. A mixture of K Gaussian distributions produces a marginal distribution whose excess kurtosis is bounded by the variance of the component means relative to the component variances. For the kurtosis of the mixture to match the observed value of 7.71, the tail states would need substantially larger emission variances or more extreme emission means than the Baum-Welch algorithm selects, but such parameters would distort the central body of the distribution and reduce KS pass rates.

The kurtosis gap explains why the AD pass rate (90.1%) is consistently lower than the KS pass rate (96.8%): the Anderson–Darling test assigns greater weight to the tails, where the Gaussian mixture’s lighter-than-observed tails produce the largest discrepancies. Replacing Gaussian emissions with Student- t distributions, whose heavier tails can produce arbitrary excess kurtosis, would address this gap while retaining the Baum-Welch estimation framework. The M-step update for Student- t emissions requires iterative estimation of the degrees-of-freedom parameter ν_k per state, adding computational cost but no conceptual difficulty, and is the natural immediate extension of this work.

Why Jumps Are Unnecessary. The discrete framework of Alswaidan and Varner [1] required a Poisson jump-duration mechanism to force the model into tail states for empirically realistic durations. The continuous model achieves comparable ACF performance without this mechanism. The explanation lies in the richer parameterization of continuous emissions.

In the discrete model, the emission distributions were estimated *separately* from the transition matrix: observations were first binned into quantile-defined states, then the transition matrix was estimated by frequency counting, and finally within-bin Student- t distributions were fitted. This three-stage procedure meant that the transition matrix was not optimized jointly with the emissions, so the learned $\mathbf{T}$ might not produce sufficient tail-state persistence, necessitating the jump mechanism as a corrective.

In the continuous model, the Baum-Welch algorithm jointly optimizes $\mathbf{T}$ and $\{(\mu_k, \sigma_k)\}$. The emission parameters and transition probabilities co-adapt: if a tail state has a distinctive emission distribution that captures extreme observations well, the posterior $\gamma_t(k)$ will assign high probability to that state during extreme events, and the M-step will increase the self-transition probability T_{kk} to reflect the clustering of such events. This joint optimization is the key advantage of the continuous approach: it allows the model to learn the persistence of tail regimes directly from data, rather than imposing it externally.

The elimination of the jump mechanism removes two hyperparameters (ϵ, λ) and the grid search required to calibrate them. For the discrete model, this grid search evaluated $6 \times 8 = 48$ parameter combinations, each requiring 200 simulated paths, for a total of 9,600 simulations per K value. The continuous model requires only the Baum-Welch iterations (20–40 to convergence), making it substantially more computationally efficient.

State Resolution and the Choice of $K = 18$. The state resolution sweep shows that K buys distributional accuracy in the 85–95% range at low cost (ACF-MAE is nearly flat), and that the gains saturate at $K = 18$. $K = 21$ is not meaningfully better than $K = 18$ on any metric, while K below 12 costs 3–6 percentage points of KS and AD pass rate relative to the $K = 18$ operating point. The practical recommendation is therefore: choose $K = 18$ for distributional-fidelity-sensitive applications (risk measurement, scenario generation), and $K = 3$ only when the downstream task is regime identification rather than synthetic generation.

The stability of ACF-MAE across K is theoretically predicted by the eigenvalue structure of $\mathbf{T}$. The dominant eigenvalue λ_2 (which controls the slowest ACF decay mode) depends on the maximum self-transition probability in the matrix. At all K values, the Baum-Welch algorithm learns at least one high-persistence central state ($T_{kk} > 0.9$, typically), ensuring that λ_2 remains close to unity regardless of the total number of states. Additional states refine the distributional

shape of the mixture but do not substantially affect the dominant eigenvalue.

Cross-Asset Robustness and OoS Degradation on Defensives. The cross-asset univariate results confirm that the CHMM generalizes to equities with diverse risk profiles (NVDA: 99.0% IS KS; AAPL: 99.6%; QQQ: 95.4%). At $K = 18$ the OoS degradation concentrates on JNJ (71.9%) and JPM (74.5%), reflecting the stationarity assumption: the CHMM’s transition matrix and emission parameters are fixed at their IS-estimated values, so regime shifts between the training and test periods, which may arise from sector-specific events or macroeconomic shocks, are not captured. This degradation is not specific to the chosen state count — the appendix sensitivity analysis shows the same pattern at every K in the sweep — so it is not a model-selection artefact but a stationarity artefact. A time-varying extension, either through rolling-window re-estimation or Bayesian online learning of the transition matrix, would likely improve OoS performance for such assets and is a natural next step.

Why Copulas Beat SIM on Cross-Asset Fidelity. The cross-asset extension tells a clean story: the two copula-based dependence models dominate the SIM factor baseline on both per-asset distributional fidelity (mean in-sample KS of 97.5% and 98.2% versus 78.4%) and cross-sectional correlation reproduction (off-diagonal MAE 0.031/0.027 versus 0.078). Two structural reasons explain this ordering.

First, rank reordering preserves marginals by construction. The copula simulation procedure in Section 3.8 never touches the per-asset simulated values except to permute them: the simulated $\hat{G}_{j,1:T}^{(p)}$ is a permutation of the stand-alone CHMM path, so its empirical CDF coincides with the CHMM marginal. The SIM, in contrast, replaces each asset’s CHMM draw with an affine combination $\alpha_j + \beta_j G_{m,t} + \eta_{j,t}$, whose distribution is a convolution of the market factor’s distribution (a CHMM mixture, rescaled by β_j) with the empirical residual distribution. This convolution does not generally coincide with the original asset’s marginal, and the mismatch grows with the distance of the asset’s factor loading from unity and with the heaviness of its idiosyncratic residuals. JPM’s 36.0% IS KS under SIM is the most dramatic illustration: JPM’s $\hat{\beta} = 1.20$ rescales an already-heavy SPY factor, and the re-injected residuals fatten the result beyond JPM’s own fitted marginal.

Second, the Student-t copula’s advantage over the Gaussian copula on correlation reproduction (0.027 vs. 0.031 off-diagonal MAE, and 0.181 vs. 0.203 Frobenius) is driven by its ability to represent joint tail dependence. The profile MLE selection of $\nu^* = 6$ is in the range ($4 \leq \nu \leq 10$) typically reported for equity-index portfolios in the risk management literature [15, 16], and implies a non-trivial lower/upper tail dependence coefficient. Concordant extreme moves across assets are substantially more likely under $C_t(\cdot; \Sigma, \nu = 6)$ than under the Gaussian copula. When the fitted correlation matrix already captures the bulk of linear co-movement, the Student-t copula’s additional tail-coupling shifts simulated correlation matrices closer to observed ones by injecting rarely-observed co-movement that the Gaussian copula never produces.

The same ordering (copulas ahead of SIM, Student-t ahead of Gaussian) is reported by Alswaidan and Varner [1] for discrete-HMM marginals. Our replication of that ordering with continuous CHMM marginals at $K = 18$ is evidence that the copula advantage is robust to the marginal family, which is a useful practical message: practitioners can swap in a richer marginal (CHMM, semi-HMM, or any other calibrated univariate model) without needing to revisit the choice of dependence structure.

Limitations. Several limitations qualify the conclusions of this study:

1. *Gaussian emission kurtosis gap.* The Gaussian assumption systematically underestimates excess kurtosis (5.1 simulated vs. 7.7 observed at $K = 18$), limiting the model’s ability to reproduce extreme tail events. Student- t or generalized hyperbolic emissions would address this.

2. *Stationarity assumption.* The transition matrix and emission parameters are estimated from the full IS window and held constant. Structural breaks (for example, the March 2020 COVID-19 crash) may not be adequately captured if the IS window does not contain comparable events. This assumption is the primary candidate explanation for the 24-percentage-point OoS KS drop on JNJ and JPM.
3. *Single OoS window.* The out-of-sample evaluation covers a single 219-day window (2025). A rolling or expanding-window evaluation would provide more robust generalization assessment and is the natural next step for the defensive-sector story.
4. *KS test i.i.d. assumption.* The two-sample KS test assumes i.i.d. observations, but each simulated path exhibits temporal dependence by construction. Volatility clustering in the simulated paths may inflate pass rates slightly relative to a block-bootstrap calibration [1].
5. *Gaussian emission symmetry.* The Gaussian emission for each state is symmetric, while empirical returns exhibit negative skewness (-0.75 for SPY). The mixture can produce overall skewness through asymmetric state occupancies and means, but may not fully capture the within-regime asymmetry.
6. *Small cross-asset universe.* The cross-asset extension is evaluated on six representative tickers, which is sufficient to establish the SIM-versus-copula ordering but does not explore scaling to the 424-asset universe of Alswaidan and Varner [1]. A full-universe study would require vine copulas or factor copulas for computational tractability and is a natural extension.
7. *No formal model selection beyond the multi-metric score.* We select $K = 18$ as the operating point on the basis of simultaneously winning all four distributional pass-rate metrics plus W_1 and simulated kurtosis, and we report the full sweep in the appendix, but we do not apply information-theoretic criteria (AIC, BIC) directly. A principled BIC/held-out-log-likelihood-based selection would confirm (or refine) the multi-metric choice.

6 Conclusion

We presented a continuous hidden Markov model (CHMM) with Gaussian emissions trained via the Baum-Welch algorithm for generating synthetic financial time series, together with two cross-asset dependence extensions for multi-asset synthesis. The central univariate finding is that the CHMM reproduces all three canonical stylized facts of equity returns (heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering) at a single moderate state resolution ($K = 18$, selected from a $\{3, 6, 9, 12, 15, 18, 21\}$ sweep by winning or tying on every distributional pass-rate metric) *without* requiring jump augmentation, discretization, or any mechanism beyond the standard HMM framework.

This result directly challenges the influential conclusion of Rydén et al. [8] that Gaussian HMMs cannot reproduce the slow decay of the autocorrelation function of absolute returns. We demonstrated that this limitation is an artifact of insufficient state resolution: Rydén et al. [8] tested only $K = 2$ – 3 states, where the geometric sojourn-time distribution cannot approximate slow ACF decay. At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to sustain persistence, achieving ACF-MAE of 0.046 – 0.051 (comparable to GARCH(1,1) at 0.047) through a sum-of-exponentials approximation that effectively mimics the flexible sojourn-time distributions of hidden semi-Markov models [9].

The direct head-to-head against the discrete framework of Alswaidan and Varner [1], reproduced as the Discrete NJ and Discrete WJ rows of Table 2, reveals a 96.8-percentage-point swing in IS KS pass rate (0% for either discrete variant $\rightarrow 96.8\%$ for the CHMM). The swing

is attributable to the move from bin-centroid emissions to continuous Gaussian emissions: once the simulated return series carries continuous-valued information, standard distributional tests on the return series pass; once it is quantized to 13 centroids, they fail by construction. The elimination of the jump mechanism further removes two hyperparameters (ϵ for jump probability, λ for mean jump duration) and the associated grid search, reducing computational cost and model complexity.

The seven-model benchmark confirmed that no single generator dominates all metrics, but the CHMM at $K = 18$ achieves the most balanced performance: 96.8% IS KS (vs. 11.5% GARCH and 0.0% for either discrete variant), ACF-MAE of 0.051 (vs. 0.047 GARCH), Wasserstein-1 of 0.099 (the smallest of any parametric model), and 100% quantile coverage. Cross-asset univariate generalization to NVDA (99.0% IS KS), JNJ (99.7%), JPM (98.4%), AAPL (99.6%), and QQQ (95.4%) confirmed adaptability to diverse equity risk profiles, and out-of-sample evaluation on held-out 2025 data showed robust generalization on SPY/NVDA/AAPL/QQQ but a sector-specific 24-percentage-point OoS KS drop on defensives (JNJ, JPM) that is invariant to the choice of K and therefore attributable to the stationarity assumption rather than model selection.

The cross-asset extensions then showed that the CHMM marginals compose cleanly with standard cross-sectional dependence models. On the six-asset universe, both the Gaussian and Student-t copulas substantially outperformed the Single Index Model factor baseline: copulas achieved a median in-sample KS pass rate of 97.8%/99.0% versus 83.8% for SIM, and an off-diagonal correlation MAE of 0.031 (Gaussian) and 0.027 (Student-t) versus 0.078 for SIM. The Student-t copula’s profile MLE selected $\nu^* = 6$ degrees of freedom, and the resulting tail-dependent copula produced the best correlation reproduction overall. These findings replicate the ordering reported by Alswaidan and Varner [1] with discrete HMM marginals, suggesting that the copula advantage is intrinsic to the rank-reordering construction and transfers across marginal families.

Extension of the framework to VIX log returns demonstrated that the CHMM captures the distinct statistical properties of volatility indices (positive skewness, mean reversion, and extreme kurtosis) and provides a natural bridge to option pricing through regime-switching geometric Brownian motion parameterized by CHMM-decoded volatility levels. A detailed comparison of the CHMM-implied volatility surface against a parametric stochastic-volatility benchmark is the subject of companion work led by our group, and intentionally lies outside the scope of this paper.

Several directions for future work follow from these results:

1. *Heavy-tailed emissions.* Replacing Gaussian emissions with Student- t or generalized hyperbolic distributions within the Baum-Welch framework would close the 2.6-unit kurtosis gap (7.71 observed vs. 5.11 simulated at $K = 18$) while retaining joint optimization of emissions and transitions. This is the single most impactful immediate extension.
2. *Time-varying transition dynamics.* Estimating the transition matrix via rolling windows or Bayesian online learning would relax the stationarity assumption and should directly address the OoS cliff observed on JNJ and JPM in 2025.
3. *Full-universe multi-asset generation via vine copulas.* Scaling the copula construction from six assets to the 424-asset universe of Alswaidan and Varner [1] requires vine or factor copulas for tractability and is a natural next step.
4. *Principled state selection.* Our $K = 18$ choice is a multi-metric score on top of a finite sweep; combining it with BIC or held-out log-likelihood (walk-forward) would yield a formally principled selection that plugs directly into the appendix Table 5.

Data and Code Availability. The simulation scripts, training and test data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository:

<https://github.com/altashly1/CHMM-paper>. The core continuous HMM implementation, the SIM module, and the Gaussian / Student-t copula modules are available in the development repository: <https://github.com/altashly1/HMM-withJumps-WIP>.

Conflict of Interest. The authors declare no competing interests.

Author Contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, implemented the Single Index Model and copula-based cross-asset extensions, conducted the empirical analysis, and generated all figures and tables. C.J. contributed to methodology review and validation. All authors edited and reviewed the final manuscript.

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A Supplementary Material

A.1 Full State-Resolution Sensitivity Table

Table 5 reports the full sweep summarized by Figure 4 in the main text. Observed excess kurtosis is 7.71 IS and 6.24 OoS at all K .

Table 5: State resolution sensitivity for SPY CHMM (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 2 in the main text. Coverage was 100% at every K in both IS and OoS windows.

K	KS (%)		AD (%)		Kurtosis		ACF-MAE	W_1	H	Cov (%)
	IS	OoS	IS	OoS	Obs	Sim				
3	89.7	92.3	81.4	83.2	7.71	3.94	0.0458	0.119	0.0790	100.0
6	93.4	92.5	87.1	84.3	7.71	5.06	0.0490	0.106	0.0782	100.0
9	92.9	91.9	83.3	84.1	7.71	3.70	0.0468	0.114	0.0779	100.0
12	94.5	93.8	86.1	86.0	7.71	4.18	0.0479	0.109	0.0767	100.0
15	94.2	93.4	88.5	85.8	7.71	4.52	0.0490	0.106	0.0766	100.0
18	95.2	95.6	88.4	90.0	7.71	5.11	0.0502	0.102	0.0771	100.0
21	94.7	95.4	88.3	88.1	7.71	4.71	0.0505	0.103	0.0774	100.0

A.2 Baum-Welch Convergence

Figure 8 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.

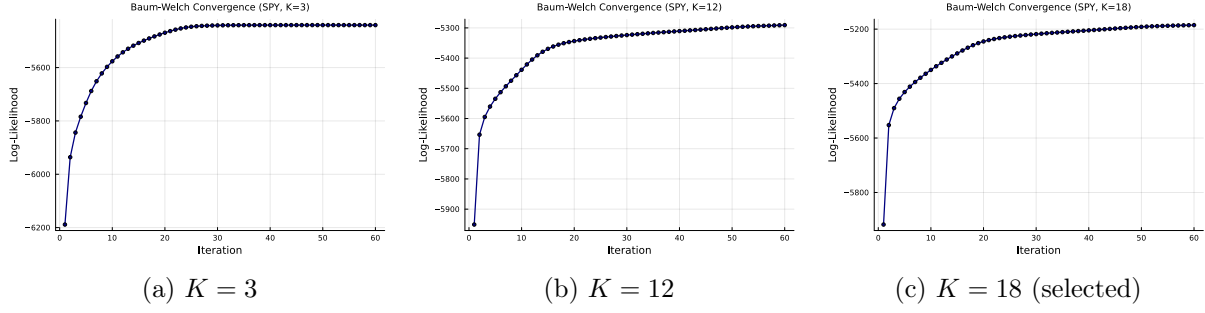


Figure 8: Baum-Welch convergence for selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` = 60.

A.3 Emission Distributions Across State Resolutions

Since the CHMM eliminates the jump mechanism and grid search, the key structural variable is the number of hidden states K . Figure 9 compares the learned emission distributions for $K \in \{3, 6, 12, 18\}$, illustrating how increasing state resolution refines the decomposition of the observed return distribution.

At $K = 3$, the model partitions the return space into three broad regimes: a crash state (large negative μ , high σ), a calm state (near-zero μ , low σ), and a boom state (positive μ , moderate σ). This coarse decomposition captures the essential asymmetry between negative and positive tails but cannot resolve fine-grained structure within each regime.

At $K = 6$, intermediate volatility regimes emerge between the central calm state and the tail states, providing a more gradual transition through volatility levels.

At $K = 12$ –18, the emission distributions form a dense coverage of the return space, with each state capturing a narrow band of returns. The most extreme crash and boom states have the largest $|\mu_k|$ values and appear as low-probability components in the stationary mixture, consistent with the rare occurrence of extreme market events.

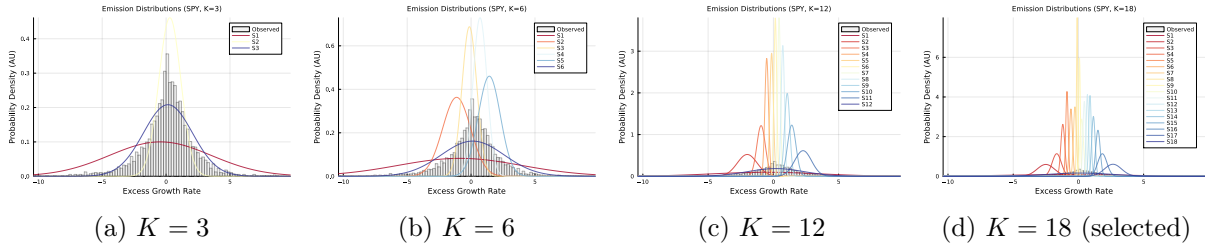


Figure 9: Learned Gaussian emission distributions across state resolutions, overlaid on the observed SPY return histogram. Higher K values produce finer decomposition with dedicated states capturing tail behavior.

A.4 Transition Matrix Structure

Figure 10 shows the learned transition matrices for $K = 6$ and $K = 18$ in $\log_{10}$ color scale. Both matrices exhibit a dominant near-diagonal band, reflecting the tendency of the market to remain in its current regime. Off-diagonal entries decay with distance from the diagonal, indicating that large regime jumps (for example, from a deep crash state directly to a boom state) are rare but nonzero.

The diagonal dominance is critical for the ACF-MAE result discussed in Section 5: the self-transition probabilities T_{kk} for central states are typically 0.7–0.95, producing mean sojourn times of 3–20 days. The mixture of these geometric sojourn times across states produces the sum-of-exponentials ACF profile that approximates the observed slow decay.

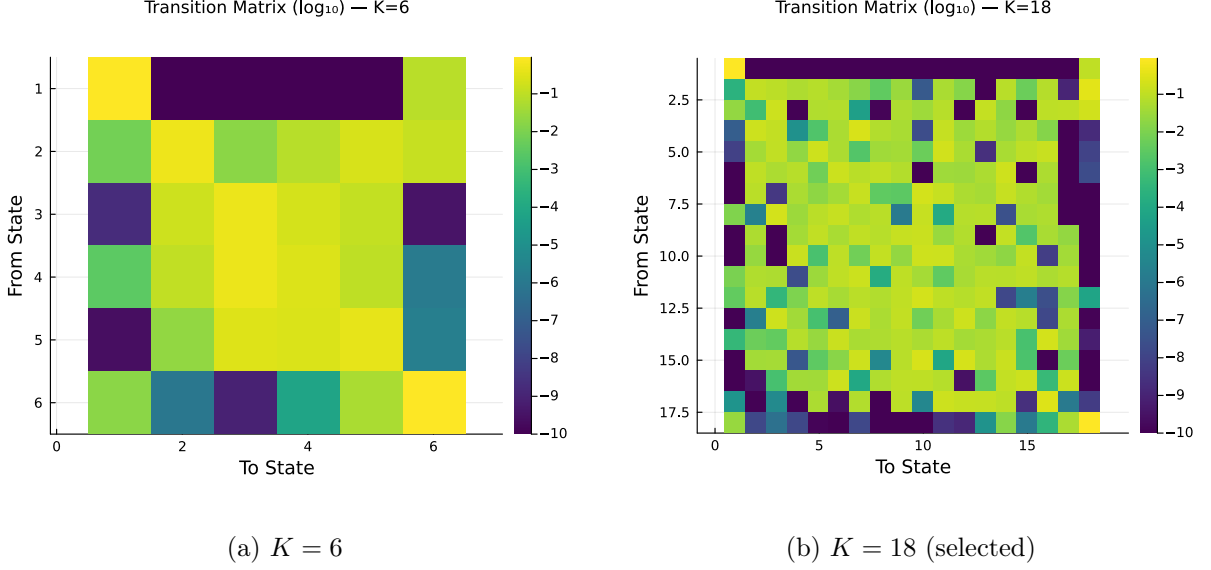


Figure 10: Learned transition matrices in $\log_{10}$ color scale. The near-diagonal band reflects regime persistence; off-diagonal entries decay with distance, indicating that large regime transitions are rare.

A.5 Stationary Distributions and Residence Times

Figure 11 shows the stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 18$. In both cases, the central states (those with μ_k near zero and moderate σ_k) receive the highest stationary probability, consistent with the empirical concentration of returns near zero. Tail states (crash and boom) receive low stationary probability, consistent with the rarity of extreme events.

Figure 12 shows the natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for $K = 6$ and $K = 18$. Under pure Markov dynamics (no jump mechanism), the expected duration in state k before transitioning is τ_k steps. Central states exhibit the longest residence times (5–20 days), while tail states have shorter residence times (1–3 days), consistent with the transient nature of extreme market events. Notably, even without a jump mechanism, the tail states at higher K maintain sufficient self-transition probability ($T_{kk} \approx 0.3$ – 0.5) to produce multi-day tail episodes. This contrasts with the discrete model of Alswaidan and Varner [1], where tail-state residence times were too short to sustain clustering without the Poisson jump extension.

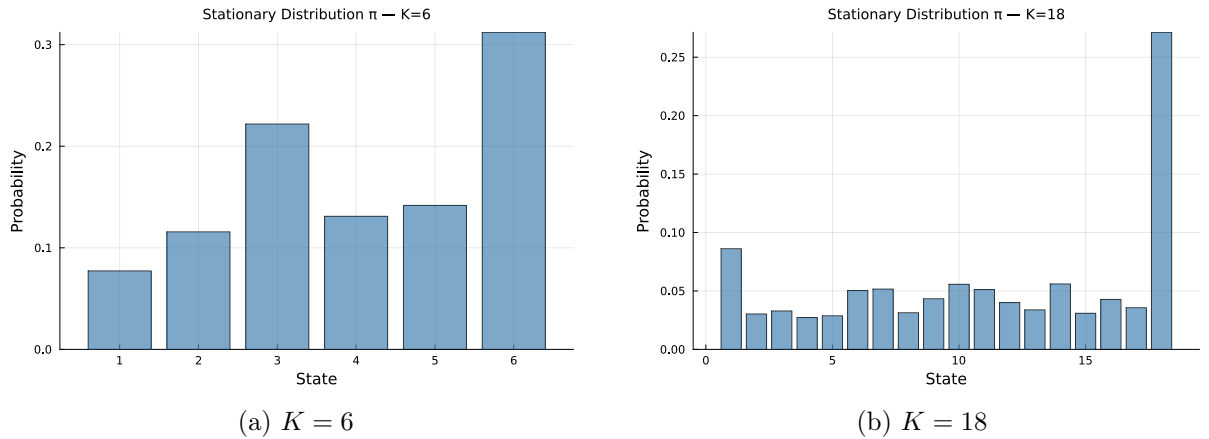


Figure 11: Stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 18$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.

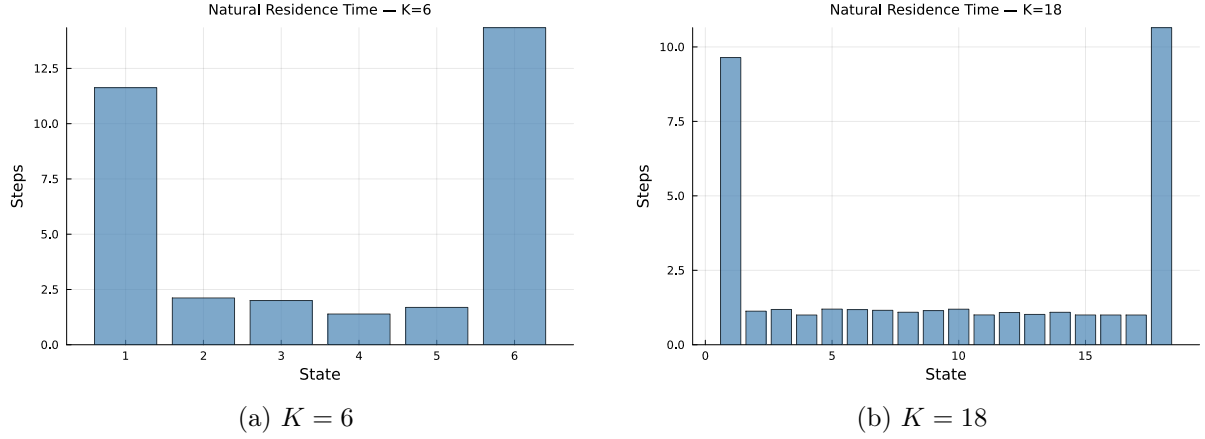


Figure 12: Natural residence times $\tau_k = 1/(1 - T_{kk})$ per state. Central states have the longest residence times (5–20 days), while tail states are shorter-lived (1–3 days). This asymmetry drives volatility clustering.

A.6 In-Sample Comparison at Non-Selected State Resolutions

Figure 13 shows the in-sample model comparison panels for $K \in \{3, 6, 9, 12, 15, 21\}$, complementing the $K = 18$ comparison in Figure 3 of the main text. Across all resolutions, the CHMM’s simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel.

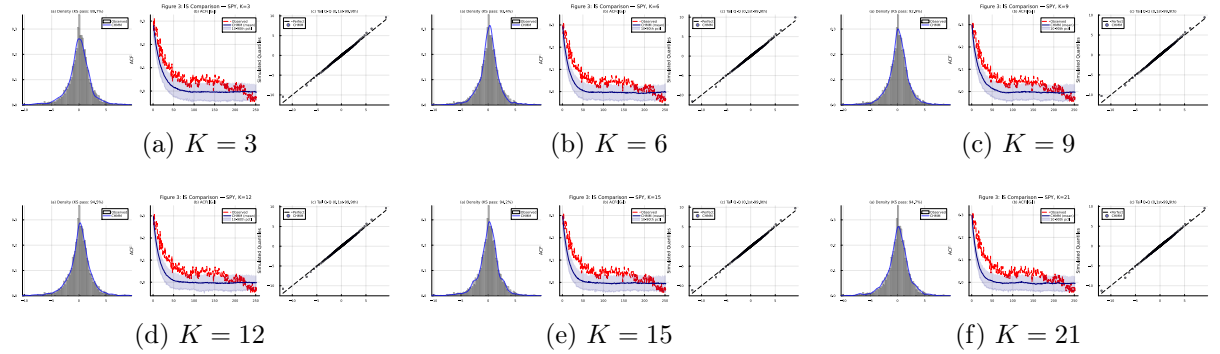


Figure 13: In-sample CHMM comparison panels at the non-selected state resolutions. Panel $K = 18$ is reproduced in Figure 3 of the main text.

A.7 Out-of-Sample Validation at Non-Selected State Resolutions

Figure 14 shows the out-of-sample validation panels for $K \in \{3, 6, 9, 12, 15, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 5 of the main text.

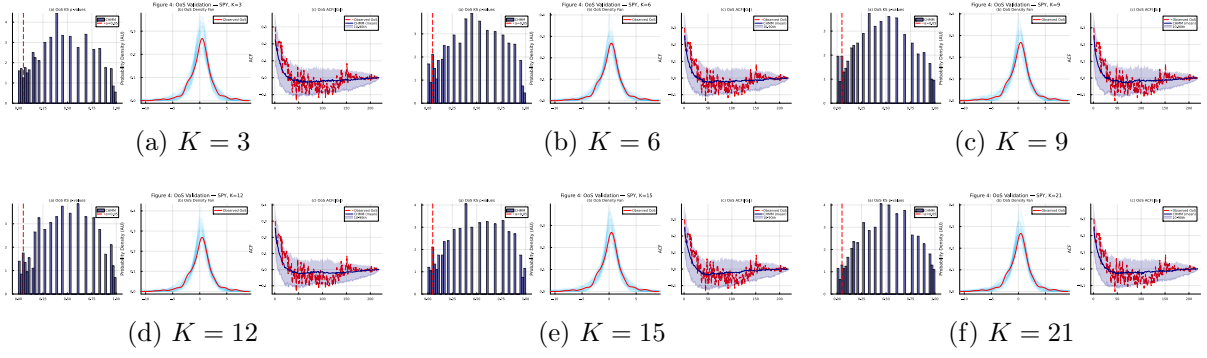


Figure 14: Out-of-sample CHMM validation panels at the non-selected state resolutions.

A.8 ACF Comparison Across State Resolutions

Figure 15 shows the ACF of $|G_t|$ for the CHMM at $K = 3$, $K = 12$, and $K = 18$, compared to the observed ACF. At all resolutions, the simulated ACF (shown as the median with 10th–90th percentile envelope across 1,000 paths) tracks the observed slow decay pattern.

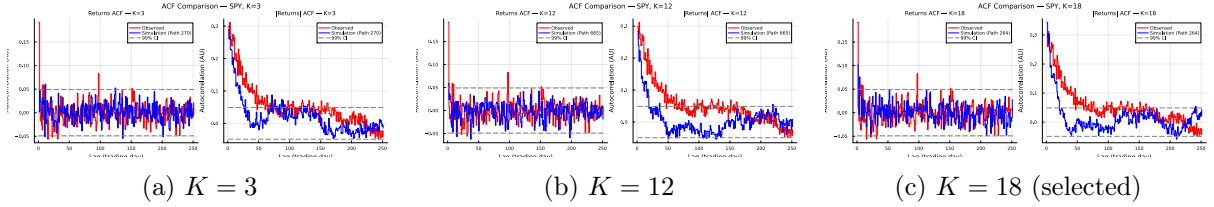


Figure 15: ACF of $|G_t|$ for the CHMM at $K = 3$, $K = 12$, and $K = 18$. The simulated 10th–90th percentile envelope (shaded) envelopes the observed ACF (red dashed), confirming that the CHMM reproduces volatility clustering without jump augmentation across the entire sweep.

A.9 Example Simulated Trajectories

Figure 16 shows example simulated return trajectories for $K = 6$ and $K = 18$, illustrating the qualitative dynamics produced by the CHMM. Both trajectories exhibit the characteristic pattern of financial returns: extended periods of low volatility punctuated by clusters of large-magnitude events.

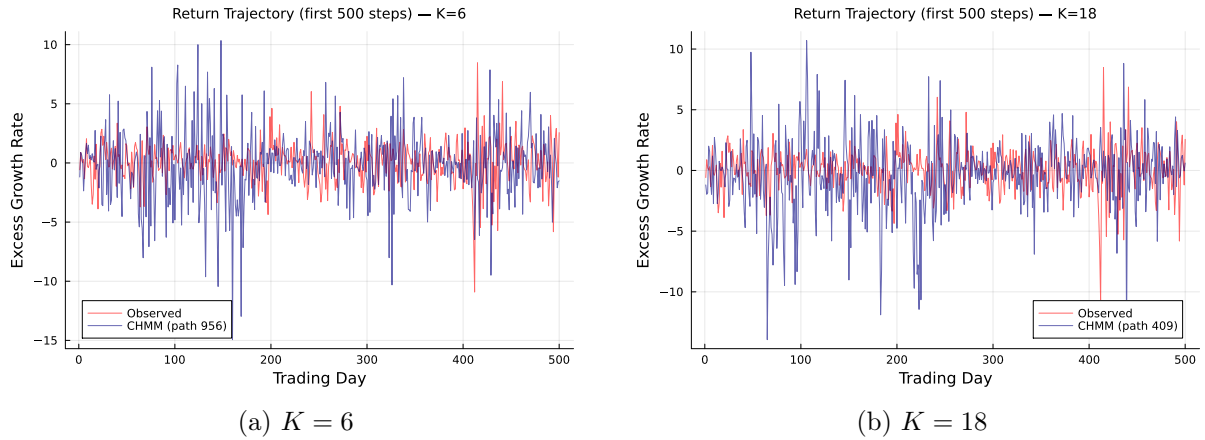


Figure 16: Example simulated return trajectories. The alternation between calm (central states) and volatile (tail states) periods produces the volatility clustering characteristic of empirical financial returns.

A.10 Cross-Asset Dependence Models: Derivations and Diagnostics

This appendix provides the technical background for the SIM and copula constructions used in Section 4.7 of the main text.

Sklar’s theorem and the rank-reordering simulator. Sklar’s theorem [13, 14] states that any d -dimensional joint CDF $H(g_1, \dots, g_d)$ with continuous marginals $F_1, \dots, F_d$ can be written as $H = C(F_1, \dots, F_d)$ for a unique copula C . The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j -th column is an i.i.d. sample from F_j , and $\mathbf{U}$ is an independent sample from C , then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$ ’s produces a sample from H , up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [27].

Kendall’s τ inversion. Given in-sample returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall’s τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [16, 33]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν . The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0, 1]^d$ is

$$\log c_t(\mathbf{u}; \Sigma, \nu) = \log \Gamma\left(\frac{\nu+d}{2}\right) + (d-1) \log \Gamma\left(\frac{\nu}{2}\right) - d \log \Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2} \log |\Sigma| - \frac{\nu+d}{2} \log \left(1 + \frac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^d \frac{\nu+1}{2} \log \left(1 + \frac{x_j^2}{\nu}\right) \quad (31)$$

where $x_j = t_\nu^{-1}(u_j)$. Summing over $t = 1, \dots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall’s- τ estimate. We evaluate equation (31) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 4.7 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d -dimensional Gaussian copula with correlation Σ , we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν , we draw $W \sim \chi_\nu^2/\nu$ independent of Y and form the multivariate- t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 6 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 4.7. QQQ’s high $R^2 = 0.841$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ’s lower $R^2 = 0.225$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation in equation (22) re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 4.

Table 6: SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\text{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014–2024 in-sample window ($T = 2,766$). Coefficients $\hat{\alpha}$, $\hat{\beta}$ and R^2 are reported for the factor equation used in equation (22).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.347	1.749	0.370
JNJ	−0.016	0.539	0.225
JPM	0.001	1.204	0.483
AAPL	0.106	1.194	0.474
QQQ	0.041	1.134	0.841

A.11 VIX Results Detail

The CHMM framework was applied to VIX daily log returns to test generalization beyond equity returns. VIX returns exhibit positive skewness (reflecting asymmetric volatility spikes), pronounced mean reversion, and extreme excess kurtosis, properties that differ fundamentally from equity returns.

The CHMM captured these properties effectively: at higher state resolutions ($K \geq 9$), the model achieved IS KS pass rates above 99%, demonstrating that the Gaussian mixture mechanism can approximate even the positively skewed, heavy-tailed marginal distribution of VIX returns. The regime decomposition provided by the Viterbi algorithm aligned with known market events, assigning high-volatility states to crisis episodes and low-volatility states to calm market periods.

The VIX extension demonstrates that the CHMM framework is not limited to equity returns but applies more broadly to financial time series with regime-switching dynamics. The decoded regimes and their associated median VIX levels provide the volatility map described in Section 3.10, linking the statistical model to the regime-switching GBM simulation.

A.12 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 describes the Baum-Welch (EM) procedure for learning CHMM parameters from continuous observations. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 5 details the SIM regression-and-resampling pipeline.

Algorithm 1 Baum-Welch (EM) algorithm for continuous Gaussian HMM parameter estimation

Require: Observations $O_{1:T}$, Number of states K , Tolerance tol , Max iterations max_iter

Ensure: Transition matrix $\mathbf{T}$, Emission parameters $\{\mu_k, \sigma_k\}_{k=1}^K$, Initial distribution $\boldsymbol{\pi}$, Log-likelihood history

```

1: Quantile-Based Initialization.
2: Sort observations:  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$ .
3: Partition  $O^{(\text{sort})}$  into  $K$  equal-sized chunks  $\{C_1, \dots, C_K\}$ .
4: for  $k = 1$  to  $K$  do
5:    $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ .
6: end for
7:  $T_{ij} \leftarrow 1/K$  for all  $i, j$ ;  $\pi_k \leftarrow 1/K$  for all  $k$ .
8:  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
9: EM Loop.
10: for  $n = 1$  to  $\text{max\_iter}$  do
11:   E-Step: Compute emission log-likelihoods  $\log B_{t,k} = \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$  for all  $t, k$ .
12:   Forward pass:
13:    $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$  for all  $k$ .
14:   for  $t = 2$  to  $T$  do
15:     for  $j = 1$  to  $K$  do
16:        $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$ 
17:     end for
18:   end for
19:   Backward pass:
20:    $\log \beta_T(k) \leftarrow 0$  for all  $k$ .
21:   for  $t = T - 1$  down to  $1$  do
22:     for  $i = 1$  to  $K$  do
23:        $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$ 
24:     end for
25:   end for
26:   Posterior quantities:
27:   for  $t = 1$  to  $T$  do
28:      $\gamma_t(k) \leftarrow \exp\left(\log \alpha_t(k) + \log \beta_t(k) - \underset{j}{\text{logsumexp}}(\log \alpha_t(j) + \log \beta_t(j))\right)$  for all  $k$ .
29:   end for
30:   for  $t = 1$  to  $T - 1$  do
31:     for  $i = 1$  to  $K$  do
32:       for  $j = 1$  to  $K$  do
33:          $\xi_t(i, j) \leftarrow \exp(\log \alpha_t(i) + \log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j) - \text{norm})$ 
34:       end for
35:     end for
36:   end for
37:   M-Step: Re-estimate parameters.
38:    $\pi_k \leftarrow \gamma_1(k)$  for all  $k$ .
39:   for  $k = 1$  to  $K$  do
40:      $\mu_k \leftarrow \sum_{t=1}^T \gamma_t(k) \cdot O_t / \sum_{t=1}^T \gamma_t(k)$ 
41:      $\sigma_k \leftarrow \max\left(\sqrt{\sum_{t=1}^T \gamma_t(k) \cdot (O_t - \mu_k)^2 / \sum_{t=1}^T \gamma_t(k)}, 10^{-6}\right)$ 
42:   end for
43:   for  $i = 1$  to  $K$  do
44:      $T_{i,j} \leftarrow \sum_{t=1}^{T-1} \xi_t(i, j) / \sum_{t=1}^{T-1} \gamma_t(i)$  for all  $j$ .
45:   end for
46:   Convergence check.
47:    $\mathcal{L}^{(n)} \leftarrow \underset{k}{\text{logsumexp}}(\log \alpha_T(k))$ .
48:   if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then
49:     break
50:   end if
51:    $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$ .
52: end for
53: return  $\mathbf{T}, \{\mu_k, \sigma_k\}_{k=1}^K, \boldsymbol{\pi}$ 

```

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$

Ensure: Most probable state sequence $S_{1:T}^*$

```
1: for  $k = 1$  to  $K$  do
2:    $\log \delta_1(k) \leftarrow \log(1/K) + \log \mathcal{N}(O_1 \mid \mu_k, \sigma_k^2)$ .
3: end for
4: for  $t = 2$  to  $T$  do
5:   for  $j = 1$  to  $K$  do
6:      $\text{vals}(j) \leftarrow \log \delta_{t-1}(j) + \log T_{ij}$  for all  $i$ .
7:      $\log \delta_t(j) \leftarrow \max_i \text{vals}(i) + \log \mathcal{N}(O_t \mid \mu_j, \sigma_j^2)$ .
8:      $\psi_t(j) \leftarrow \arg \max_i \text{vals}(i)$ .
9:   end for
10: end for
11:  $S_T^* \leftarrow \arg \max_k \log \delta_T(k)$ .
12: for  $t = T - 1$  down to  $1$  do
13:    $S_t^* \leftarrow \psi_{t+1}(S_{t+1}^*)$ .
14: end for
15: return  $S_{1:T}^*$ 
```

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M , Number of paths P

Ensure: Simulated return paths $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$

```
1: for  $p = 1$  to  $P$  do
2:   Sample initial state  $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ .
3:   for  $t = 1$  to  $M$  do
4:     Emit observation  $\hat{G}_t^{(p)} \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$ .
5:     if  $t < M$  then
6:       Transition to next state  $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ .
7:     end if
8:   end for
9: end for
10: return  $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$ 
```

Algorithm 4 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (PSD, unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

```

1: for  $p = 1$  to  $P$  do
2:   Draw copula sample  $\mathbf{U}^{(p)} \in [0, 1]^{T \times d}$  from  $\mathcal{C}(\Sigma)$ :
3:    $L \leftarrow \text{Cholesky}(\Sigma)$ ,  $Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$  i.i.d.
4:    $Y \leftarrow ZL^\top$ 
5:   if  $\mathcal{C} = \text{Gaussian}$  then
6:      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$  componentwise
7:   else
8:     Draw  $W \sim \chi_\nu^2/\nu$  i.i.d. for  $t = 1, \dots, T$ ;  $X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$ 
9:      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$  componentwise
10:  end if
11:  for  $j = 1$  to  $d$  do
12:    Simulate  $\tilde{g}_{j,1:T}^{(p)}$  from  $\mathcal{M}_j$  via Algorithm 3.
13:    Compute order statistics  $\tilde{g}_{j,(1)}^{(p)} \leq \dots \leq \tilde{g}_{j,(T)}^{(p)}$ .
14:    Compute ranks  $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$  for  $t = 1, \dots, T$ .
15:     $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$  for  $t = 1, \dots, T$ .
16:  end for
17: end for
18: return  $\{\hat{G}_{j,t}^{(p)}\}$ 

```

Algorithm 5 Single Index Model (SIM) construction and simulation

Require: In-sample return matrix $\mathbf{R} \in \mathbb{R}^{T \times d}$, market-column index m , market CHMM $\mathcal{M}_m$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

```

1: OLS fit. For  $j \neq m$ :
2:    $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m}) / \text{Var}(\mathbf{R}_{:,m})$ 
3:    $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$ 
4:    $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}$ ,  $R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$ 
5: for  $p = 1$  to  $P$  do
6:   Simulate market path  $G_{m,1:T}^{(p)}$  from  $\mathcal{M}_m$  via Algorithm 3.
7:    $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$ .
8:   for  $j \neq m$  do
9:     Draw residual indices  $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \dots, T\})$ .
10:     $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$ .
11:  end for
12: end for
13: return  $\{\hat{G}_{j,t}^{(p)}\}$ 

```
